Does a protective skin microbiome save the population?

Building biological realism one step at a time

Table of contents

The question and the method . . . 1
Part A. Bsal in fire salamanders . . . 3
  Host and pathogen . . . 3
  Treatment . . . 9
  Part A summary . . . 18
Part B. Bd in common toads . . . 18
  Host and pathogen . . . 19
  Treatment . . . 27
  Part B summary . . . 31
Part C. Bd in frogs . . . 31
  Host and pathogen . . . 31
  Treatment . . . 35
  Part C summary . . . 41
How much did each step move the answer? . . . 41
Does the effect of one piece of realism depend on the others? . . . 43
All steps . . . 45
What the exercise shows . . . 49
What is not in the models . . . 50
References . . . 51

The question and the method

Amphibians can be treated with skin bacteria that inhibit the chytrid fungi *Batrachochytrium dendrobatidis* (Bd) and *B. salamandrivorans* (Bsal). Would that protect a population?

The answer a model gives depends on how much biology is in it. This vignette makes that dependence visible. For each of three host–pathogen systems it:

1. starts from the simplest model of the host and the pathogen;

2. adds the biology of the host and pathogen one piece at a time, with no treatment;
3. adds the biology of the treatment one piece at a time;
4. after every step, records the same two outcomes: the population without treatment, and the population under one fixed reference treatment;
5. ends with chance: the last model of each part is run again with whole animals and random events, many times over.

Most steps add one thing, and the step before is recovered by switching it off. Where a step instead swaps a mechanism or sets an earlier addition aside, it says so; the equations at the start of each part list the settings every step uses. The final sections measure how much each addition moved the answer, and check with a factorial design whether the effect of one piece of realism depends on the others. A second vignette repeats the exercise in a single nested model, where every combination of pieces can be compared like for like.

Part	System	Simplest model
A	Bsal in fire salamanders	susceptible and infectious adults
B	Bd in common toads	Mitchell et al. (2008) pond model
C	Bd in frogs	Briggs et al. (2010) load model

> **!** Important
>
> The published models (Schmidt et al. 2017, Mitchell et al. 2008, Briggs et al. 2010) are reproduced from their papers. Everything added to them is new work in this package and is not a published result. Values taken from experiments are read from the package's parameter compendium and are named where they are used. All other added values (efficacy, coverage, spread, exposure windows, clearance) are illustrative: they show what the model does, not what a real treatment would achieve.

```
using Chytridiomycosis, Plots, Printf
using Distributions: LogNormal, quantile
using Random: Xoshiro
default(linewidth = 2, size = (720, 380), legendfontsize = 8, guidefontsize
 ↪  = 9,
    titlefontsize = 10, left_margin = 4Plots.mm, bottom_margin = 4Plots.mm)

include("../diagram.jl")
T(; kw...) = MicrobiomeTreatment(; kw...)

# One row per step, filled in as we go. `untreated` and `treated` are the
 ↪  two outcomes;
# NaN where one does not apply.
ladder = NamedTuple[]
function step!(system, phase, added, untreated, treated, note)
    n = count(r -> r.system == system, ladder)
```

```
    push!(ladder, (; system, n, phase, added, untreated, treated, note))
    return (step = n, untreated = round(untreated, digits = 3), treated =
     ↪  round(treated, digits = 3))
end

function ladder_plot(system, xlabel)
    rows = filter(r -> r.system == system, ladder)
    k = length(rows)
    labels = ["$(r.n). $(r.added)" for r in rows]
    z(x) = isnan(x) ? 0.0 : x
    bar((1:k) .- 0.2, [z(r.untreated) for r in rows]; orientation = :h,
     ↪  bar_width = 0.38,
        label = "no treatment", color = :gray70, yticks = (1:k, labels),
 ↪  yflip = true, xlabel,
        xlims = (0, 1.05), xticks = 0:0.25:1, size = (840, 80 + 36k),
 ↪  left_margin = 6Plots.mm,
        legend = :outertop, legend_column = 2)
    bar!((1:k) .+ 0.2, [z(r.treated) for r in rows]; orientation = :h,
     ↪  bar_width = 0.38,
        label = "with treatment", color = :steelblue)
end
```

```
ladder_plot (generic function with 1 method)
```

Part A. Bsal in fire salamanders

The outcome in Part A is the mean number of adults over the ten years after Bsal arrives, as a fraction of carrying capacity. The reference treatment is: every adult treated once, on the day Bsal arrives, with bacteria that block 99% of infections for life.

Host and pathogen

Step 0. Susceptible and infectious adults

The simplest model: an adult is susceptible or infectious, infection passes by contact, and infectious animals die of the disease. This is the model of Schmidt et al. (2017) with its latent period made negligibly short.

```
p = SchmidtParameters(:schmidt2017)
p_si = Chytridiomycosis.with(p; e = 5000.0)          # latent period under
 ↪  two hours
block = T(efficacy = 0.99)
```

```
sli(q, tr; kw...) = solve(ProbioticSLIProblem(q, tr; kw...))
u0, t0 = sli(p_si, T()), sli(p_si, block; coverage = 1.0)
@printf("R0 = %.1f\n", basic_reproduction_number(p_si))
step!("Bsal", "host", "susceptible and infectious adults", u0.mean_hosts,
 ↪  t0.mean_hosts,
    "collapse; treated adults live but are not replaced")
```

```
R0 = 24.8
```

```
(step = 0, untreated = 0.003, treated = 0.338)
```

Step 1. A latent period

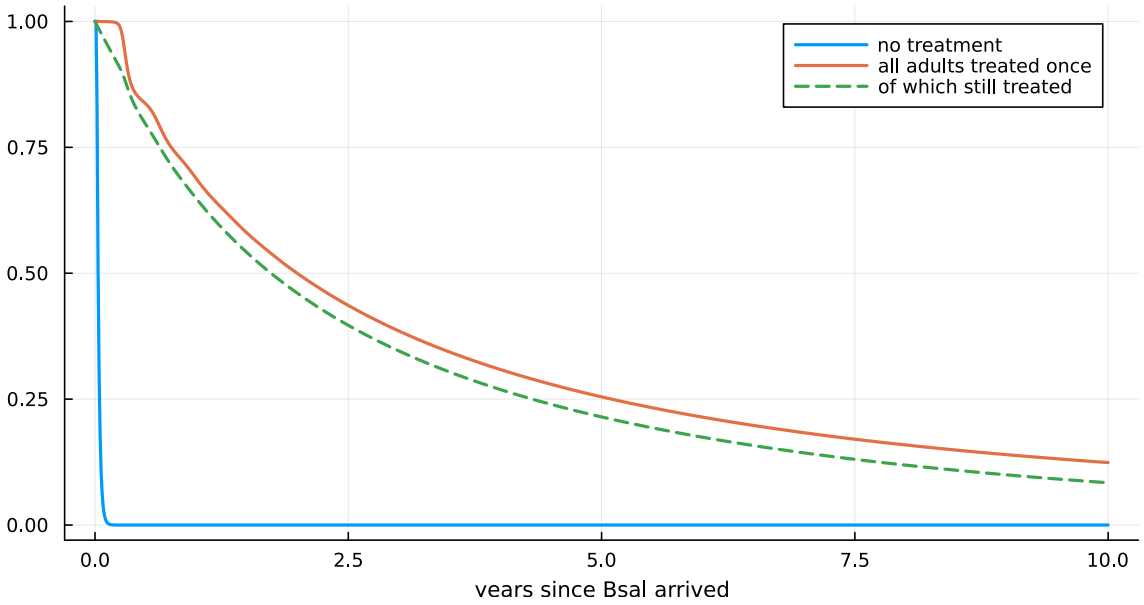
Infected animals are not infectious at once (Schmidt et al. 2017, as published).

```
u1, t1 = sli(p, T()), sli(p, block; coverage = 1.0)
@printf("R0 = %.1f\n", basic_reproduction_number(p))
step!("Bsal", "host", "+ latent period (Schmidt 2017)", u1.mean_hosts,
 ↪  t1.mean_hosts,
    "almost no change: a one-day delay against a one-week illness")
```

```
R0 = 24.8
```

```
(step = 1, untreated = 0.004, treated = 0.338)
```

```
plot(u1.t, u1.hosts ./ p.K; label = "no treatment", xlabel = "years since
 ↪  Bsal arrived",
    ylabel = "adults (fraction of K)")
plot!(t1.t, t1.hosts ./ p.K; label = "all adults treated once")
plot!(t1.t, t1.treated ./ p.K; label = "of which still treated", linestyle =
 ↪  :dash)
```



Treated adults survive, but the population still declines: recruits are born untreated and Bsal is still circulating.

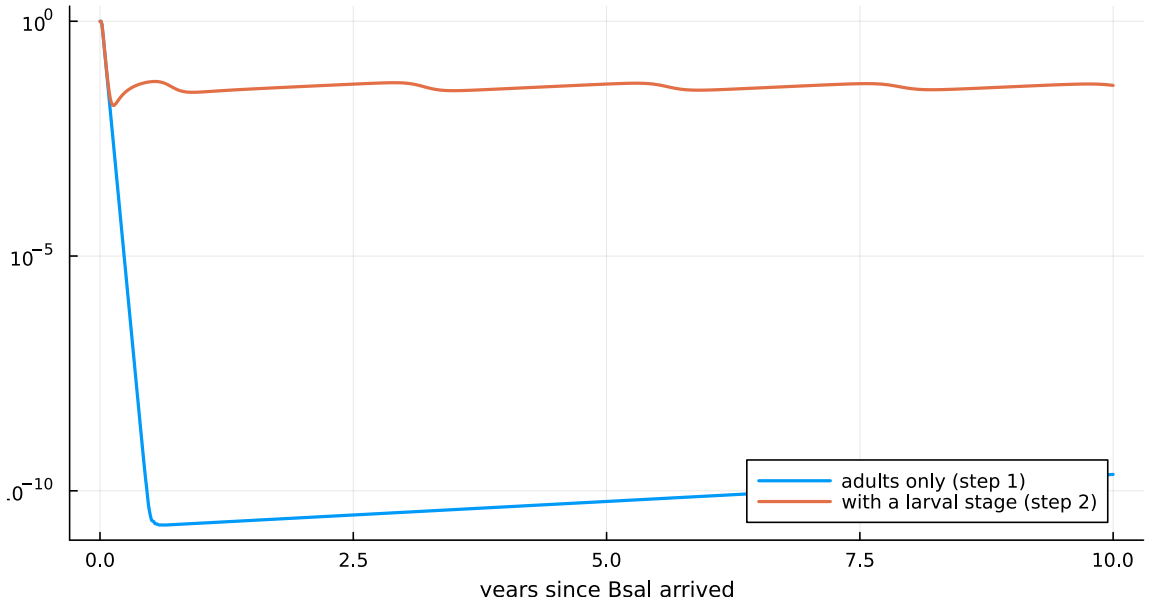
Step 2. A larval stage

Salamanders have aquatic larvae. Here they are added as a stage that Bsal does not infect.

```
clean_larvae = BsalLarvalStage(carry_over = 0.0)
lar(st, tr; kw...) = solve(ProbioticLarvalProblem(p, st, tr; kw...))
u2, t2 = lar(clean_larvae, T()), lar(clean_larvae, block; coverage = 1.0)
step!("Bsal", "host", "+ larval stage, not infected", u2.mean_adults,
 ↪  t2.mean_adults,
    "untreated population no longer vanishes: larvae are a refuge")
```

```
(step = 2, untreated = 0.044, treated = 0.336)
```

```
plot(u1.t, max.(u1.hosts ./ p.K, 1e-12); yscale = :log10, label = "adults
 ↪  only (step 1)",
    xlabel = "years since Bsal arrived", ylabel = "adults (fraction of K),
 ↪  no treatment",
    legend = :bottomright)
plot!(u2.t, u2.adults ./ p.K; label = "with a larval stage (step 2)")
```



Larvae already in the water metamorphose after the adult epidemic has burnt out. The near-extinction in the adults-only model was partly a result of having no other stage.

Step 3. Larvae can be infected

Fire salamander larvae can be infected experimentally without showing disease, and half carried the infection through metamorphosis (Wang et al. 2021; the count is read from the compendium). The transmission rate from adults to larvae is illustrative.

```
stage = BsalLarvalStage(:wang2021; adult_to_larva = 5.2)
@printf("carry-over through metamorphosis: %.1f; R0 = %.1f\n",
 ↪  stage.carry_over,
    basic_reproduction_number(p, stage))
u3, t3 = lar(stage, T()), lar(stage, block; coverage = 1.0)
step!("Bsal", "host", "+ larvae infected, carry-over 0.5", u3.mean_adults,
 ↪  t3.mean_adults,
    "small change at this carry-over")
```

```
carry-over through metamorphosis: 0.5; R0 = 24.9
```

```
(step = 3, untreated = 0.039, treated = 0.334)
```

This is the host model used for the rest of Part A.

The host model in equations

Densities are adult females and larvae per hectare; time is in years. Steps 0 and 1 use `ProbioticSLIProblem`, which has adults only. Step 1 is the model of Schmidt et al. (2017):

$$\begin{aligned}\frac{dS}{dt} &= bS - (d + \gamma N)S - \beta SI \\ \frac{dL}{dt} &= \beta SI - (d + e + \gamma N)L \\ \frac{dI}{dt} &= eL - (d + d_I + \gamma N)I\end{aligned}$$

with $N = S + L + I$ and $\gamma = (b - d)/K$, so that without Bsal S settles at K. Only susceptible adults breed. Bsal enters as $I = 0.01$ per hectare (one per square kilometre) at $t = 0$.

- Step 0 is the same system with $e = 5000$ per year, so that L empties within hours and the model is in effect $S \to I$. The printed $R_0 = \beta eK/((b + e)(b + d_I))$, which tends to $\beta K/(b + d_I)$ as e grows.
- Step 1 sets e to its published value: a latent period of about one day.

Steps 2 and 3 use `ProbioticLarvalProblem`, which adds uninfected and infected larvae, W_S and W_I:

$$\begin{aligned}\frac{dW_S}{dt} &= b'S - (m + \mu_W)W_S - \beta_{WA} I W_S \\ \frac{dW_I}{dt} &= \beta_{WA} I W_S - (m + \mu_W)W_I \\ \frac{dS}{dt} &= mW_S + (1 - c)\, mW_I - (d + \gamma N)S - \beta SI \\ \frac{dL}{dt} &= c\, mW_I + \beta SI - (d + e + \gamma N)L \\ \frac{dI}{dt} &= eL - (d + d_I + \gamma N)I\end{aligned}$$

- Step 2 replaces the recruitment term bS in dS/dt with births into the larval stage, $b'S$ with $b' = b(m + \mu_W)/m$. Larvae leave by metamorphosis at rate m or die at rate μ_W. The choice of b' keeps the Bsal-free state of step 1 ($S = K$, plus $W_S = bK/m$ larvae). Here $\beta_{WA} = 0$ and $c = 0$, so W_I stays at zero.
- Step 3 adds infection of larvae by infectious adults ($\beta_{WA} I W_S$) and carry-over: an infected larva becomes a latent adult with probability c, and a susceptible adult otherwise. Infected larvae do not die of Bsal and, in Part A, infect no one. (The code also has larva-to-larva and larva-to-adult transmission, `larva_to_larva` and `larva_to_adult`; both are zero throughout Part A.) The printed R_0 is the largest eigenvalue of the next-generation matrix for (W_I, L, I) at the Bsal-free state.

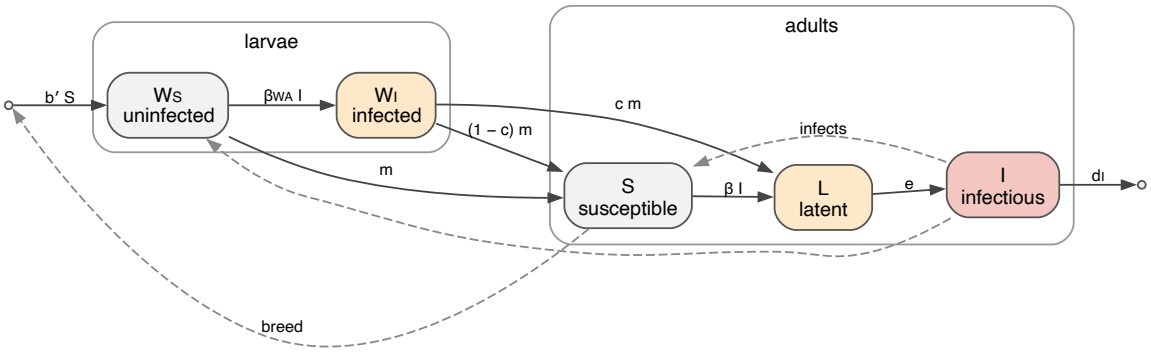


Figure 1: The host model after step 3. Solid arrows are flows, labelled with the per-capita rate; dashed arrows show who breeds and who infects whom. Not drawn: every adult class also dies at rate d + γN and every larval class at the larval death rate.

Symbol	Meaning	Value or source
S, L, I	susceptible, latent and infectious adults (per ha)	start at $S = K$
W_S, W_I	uninfected and infected larvae (per ha)	start at $W_S = bK/m$
b	recruitment rate	0.39 per yr, compendium (schmidt2017)
d	natural death rate	0.125 per yr, compendium (schmidt2017)
d_I	death rate from Bsal	52 per yr, compendium (schmidt2017)
e	latent to infectious	364 per yr, compendium (schmidt2017); 5000 in step 0
β	transmission between adults	52 ha per yr, compendium (schmidt2017)
K	carrying capacity	25 per ha, compendium (schmidt2017)
γ	density dependence	$(b - d)/K$
m	metamorphosis rate, 1/larval period	1/0.33 per yr, illustrative
μ_W	larval death rate	1 per yr, illustrative
b'	larvae born per susceptible adult	$b(m + \mu_W)/m$
β_{WA}	transmission from adults to larvae	0 in step 2; 5.2 ha per yr in step 3, illustrative
c	carry-over through metamorphosis	0 in step 2; 0.5 in step 3, compendium (wang2021: 5 of 10 larvae)

The outcome is the mean of $(S + L + I)/K$ over the ten years after Bsal arrives.

Treatment

The reference treatment of step 3 is now made more realistic. Each step states what changed.

The treatment in equations

Every class is split into untreated (U) and treated (T). With all treatment terms present, the model of `ProbioticLarvalProblem` is, for adults,

$$\begin{aligned}
\frac{dS_U}{dt} &= m\,[W_{SU} + (1-r)W_{ST}] + (1-c)\,m\,[W_{IU} + (1-r)W_{IT}] \\
&\quad - (d + \gamma N)S_U - \lambda_A S_U + \omega S_T - a_A S_U \\
\frac{dS_T}{dt} &= r\,mW_{ST} + (1-c)\,r\,mW_{IT} - (d + \gamma N)S_T - (1-\varepsilon)\lambda_A S_T - \omega S_T + a_A S_U \\
\frac{dL_U}{dt} &= c\,m\,[W_{IU} + (1-r)W_{IT}] + \lambda_A S_U - (d + e + \gamma N)L_U + \omega L_T - a_A L_U \\
\frac{dL_T}{dt} &= c\,r\,mW_{IT} + (1-\varepsilon)\lambda_A S_T - (d + e + \gamma N)L_T - \omega L_T + a_A L_U \\
\frac{dI_U}{dt} &= eL_U - (d + d_I + \gamma N)I_U + \omega I_T - a_A I_U \\
\frac{dI_T}{dt} &= eL_T - (d + (1-\tau)d_I + \gamma N)I_T - \omega I_T + a_A I_U
\end{aligned}$$

and for larvae

$$\begin{aligned}
\frac{dW_{SU}}{dt} &= b'\,[S_U + S_T + \tau(L_T + I_T)] - (m + \mu_W)W_{SU} - \lambda_W W_{SU} + \omega W_{ST} - a_W W_{SU} \\
\frac{dW_{ST}}{dt} &= -(m + \mu_W)W_{ST} - (1-\varepsilon)\lambda_W W_{ST} - \omega W_{ST} + a_W W_{SU} \\
\frac{dW_{IU}}{dt} &= \lambda_W W_{SU} - (m + \mu_W)W_{IU} + \omega W_{IT} - a_W W_{IU} \\
\frac{dW_{IT}}{dt} &= (1-\varepsilon)\lambda_W W_{ST} - (m + \mu_W)W_{IT} - \omega W_{IT} + a_W W_{IU}
\end{aligned}$$

with N all adults and

$$\begin{aligned}
&\lambda_A = \beta\,[I_U + (1-\sigma)I_T], \qquad \lambda_W = \beta_{WA}\,[I_U + (1-\sigma)I_T], \\
&a_A = \eta\,\frac{S_T + L_T + I_T}{N}, \qquad a_W = \eta\,\frac{W_{ST} + W_{IT}}{W_{SU} + W_{ST} + W_{IU} + W_{IT}}, \\
&\omega = \frac{\ln 2}{h}, \qquad (\varepsilon, \tau, \sigma) = g(t - t_k)\,(\varepsilon_0, \tau_0, \sigma_0), \qquad g(a) = \frac{1 + 1/F}{1 + 2^{a/h_B}/F}.
\end{aligned}$$

Treatment itself is a pulse. At each treatment time t_k, a fraction ϕ (`coverage`) of every untreated class of the treated stage moves to the matching treated class, and the clock of g restarts (t_k is the most recent application). Bsal arrives just after any treatment at the same time. Recruits are born untreated. In steps 0 and 1 (`ProbioticSLIProblem`, no larvae) the adult equations are the same without the metamorphosis terms, and recruits $b\,[S_U + S_T + \tau(L_T + I_T)]$ enter S_U directly. Treated, infected adults breed in proportion to τ: this is an assumption of the code, and with $\tau = 0$ it is Schmidt's rule. (The code also has an environmental route of spread, `environmental`, and inheritance from treated mothers, `maternal`; both are zero throughout Part A and are left out above.)

What each step switches on:

- Steps 0–3, reference treatment: `efficacy = 0.99`, adults treated once at $t = 0$ with $\phi = 1$. So $\varepsilon_0 = 0.99$ and $\omega = \eta = \tau_0 = \sigma_0 = 0$, $g = 1$: treated adults are infected at 1% of the untreated rate, for life.
- Step 4, `half_life = 1.0`: $\omega = \ln 2$ per year; treated animals revert to untreated.
- Step 5, `direct = 50.0`: $\eta = 50$ per year. An untreated animal acquires the bacteria in proportion to the treated fraction of its own stage. The vignette prints the microbiome's reproduction number $R_m = \eta/(\omega + b)$, with adult turnover b, and the long-run coverage $1 - 1/R_m$. Larvae are never treated in this step, so recruits acquire the bacteria only from adults.
- Step 6, `tolerance = 0.9` and no `efficacy`: $\varepsilon_0 = 0$, $\tau_0 = 0.9$. A treated, infected adult dies of Bsal at $(1-\tau)d_I$, stays infectious, and breeds at τ. The printed reproduction number, with all adults treated, is $R_0\,[1 - \phi + \phi M]$ with $M = (1-\varepsilon)(1-\sigma)(b + d_I)/(b + (1-\tau)d_I)$. This uses the adults-only model and ignores gain or loss of the bacteria during an infection.
- Step 7, `load_reducing_treatment` with reduction in load ρ and `death_exponent` k: $\sigma_0 = \rho$ and $\tau_0 = 1 - (1-\rho)^k$, with $\varepsilon_0 = 0$. The run uses $\rho = 0.99$, $k = 1$, so $\sigma_0 = \tau_0 = 0.99$; loss and spread as in step 5.
- Step 8: back to $\varepsilon_0 = 0.99$ with $\omega = \eta = 0$. `treat = :larvae` or `:both` applies the pulse to $W_{SU} \to W_{ST}$ and $W_{IU} \to W_{IT}$ as well, every quarter; `retention` is r. Efficacy protects treated larvae too, through $(1-\varepsilon)\lambda_W$.
- Step 9, `abundance_half_life = 5/365`, `protective_fold = 100`: g is no longer 1. It multiplies ε_0, τ_0 and σ_0 (here only ε_0 is non-zero), falls to one half about $\log_2 F$ abundance half-lives after each application, and is reset to 1 by the next. All treated animals share one clock. `half_life` is left at its default, so $\omega = 0$.

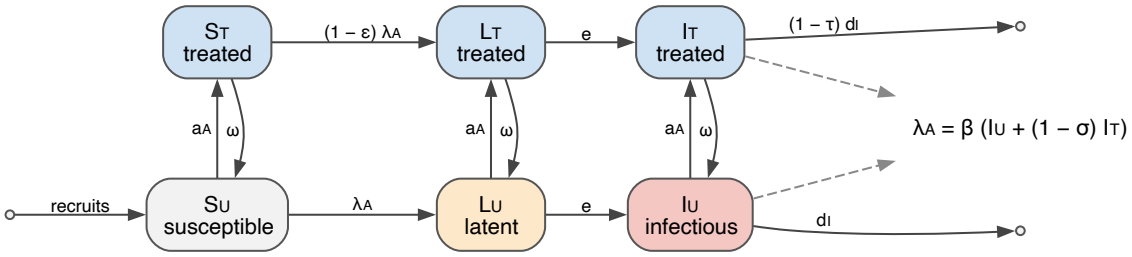


Figure 2: Treatment of adults. Blue boxes are treated animals. Solid arrows are flows with per-capita rates; dashed arrows feed the force of infection. Not drawn: background death $d + \gamma N$ of every class, and the pulses of treatment that move a fraction φ from each untreated class up to the treated one.

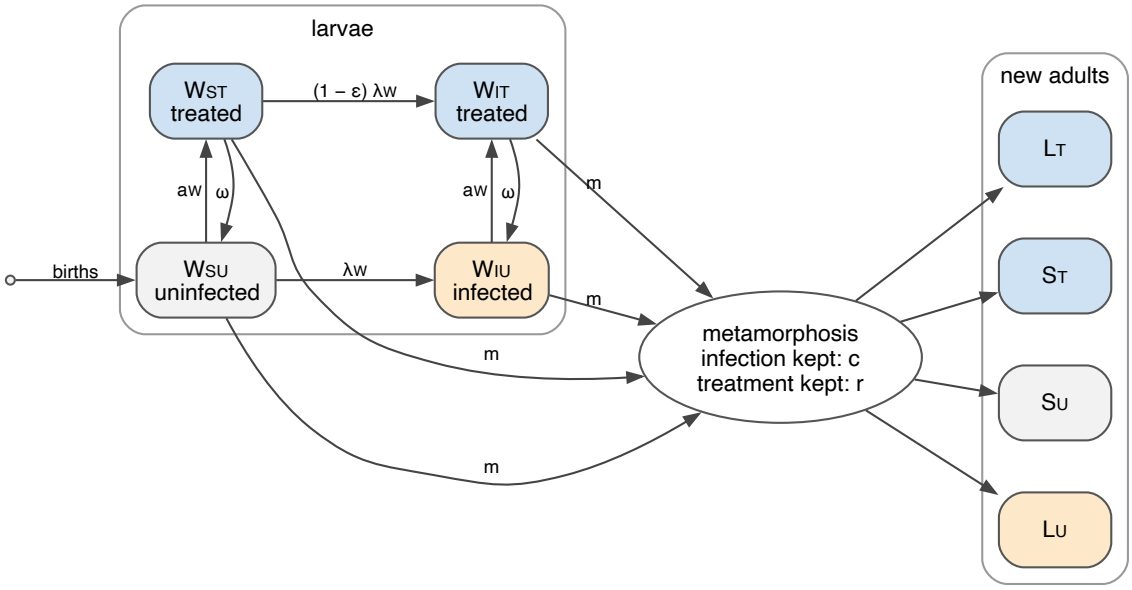


Figure 3: Treatment of larvae and what happens at metamorphosis. All larvae are born uninfected and untreated, to susceptible adults and, in proportion to τ, to treated infected adults. The force of infection on larvae has the same form as on adults, with the adult-to-larva rate in place of β. Not drawn: larval death.

Symbol	Meaning	Value or source
U, T	untreated, treated	
ε_0	`efficacy`: reduction in the infection rate	0.99 in steps 0–5, 8, 9; 0 in steps 6–7; illustrative
τ_0	`tolerance`: reduction in death from Bsal	0.9 in step 6; 0.99 in step 7; illustrative

Symbol	Meaning	Value or source
σ_0	`shedding_reduction`: reduction in shedding	0.99 in step 7; illustrative
h, ω	`half_life` of treatment and loss rate $\ln 2/h$	$h = 1$ yr in steps 4–7, otherwise none ($\omega = 0$); illustrative
η	`direct`: spread of the bacteria	50 per yr in steps 5–7, otherwise 0; illustrative
ϕ	`coverage` at each treatment	1
t_k	`treat_times`	0 in steps 0–7; every quarter from 0 to 9.75 yr in steps 8–9 (monthly in one run)
r	`retention` at metamorphosis	0 or 1 in step 8, 1 in step 9; illustrative, not measured
h_B	`abundance_half_life` of the applied bacteria	5 days in step 9, from this project's fits to Knapp et al. (2022) and Madison et al. (2025)
F	`protective_fold`	100 in step 9 (10 and 1000 in the plot); illustrative, not measured
g	`protection`: share of the full effect left after time a	1 except in step 9

Step 4. Protection is lost

A treated animal reverts to untreated, with a half-life of one year.

```
a = lar(stage, T(efficacy = 0.99, half_life = 1.0); coverage = 1.0)
step!("Bsal", "treatment", "+ protection lost, half-life 1 year",
 ↪  u3.mean_adults, a.mean_adults,
    "collapse only delayed")
```

```
(step = 4, untreated = 0.039, treated = 0.133)
```

Step 5. The bacteria spread between animals

Treated animals pass the bacteria to others, so protection can reach the recruits.

```
spread = T(efficacy = 0.99, half_life = 1.0, direct = 50.0)
@printf("microbiome reproduction number %.0f, long-run coverage %.2f\n",
    microbiome_reproduction_number(spread; turnover = p.b),
     ↪  equilibrium_coverage(spread; turnover = p.b))
```

```
a = lar(stage, spread; coverage = 1.0)
step!("Bsal", "treatment", "+ bacteria spread between animals",
 ↪  u3.mean_adults, a.mean_adults,
    "full protection: the microbiome maintains itself")
```

```
microbiome reproduction number 46, long-run coverage 0.98
```

```
(step = 5, untreated = 0.039, treated = 0.999)
```

A model stopped here says: a self-spreading, strongly protective microbiome works, and nothing else does.

Step 6. What trials show: survival, not fewer infections

In challenge trials, treated animals were infected as often as untreated ones, but fewer died. The compendium holds the counts from one trial in *Rana muscosa* (Harris et al. 2009):

```
h = parameter_set(:harris2009ismej)
mortality_reduction(h.deaths_treated, h.n_treated, h.deaths_control,
 ↪  h.n_control; rng = Xoshiro(1))
```

```
(median = 0.9528399509303813, lower = 0.5571631729768851, upper = 0.9998974605554769)
```

So the mechanism is changed: treatment reduces disease-induced death (`tolerance`) and does not block infection. A treated, infected animal stays infectious. Loss and spread are as in step 5.

```
tolerant = T(tolerance = 0.9, half_life = 1.0, direct = 50.0)
@printf("Bsal reproduction number if everyone is treated: %.0f (untreated
 ↪  %.0f)\n",
    effective_reproduction_number(p, tolerant, 1.0),
     ↪  basic_reproduction_number(p))
a = lar(stage, tolerant; coverage = 1.0)
step!("Bsal", "treatment", "mechanism as in trials: survive, still
 ↪  infected", u3.mean_adults,
    a.mean_adults, "fails; survivors shed for longer, so transmission
 ↪  rises")
```

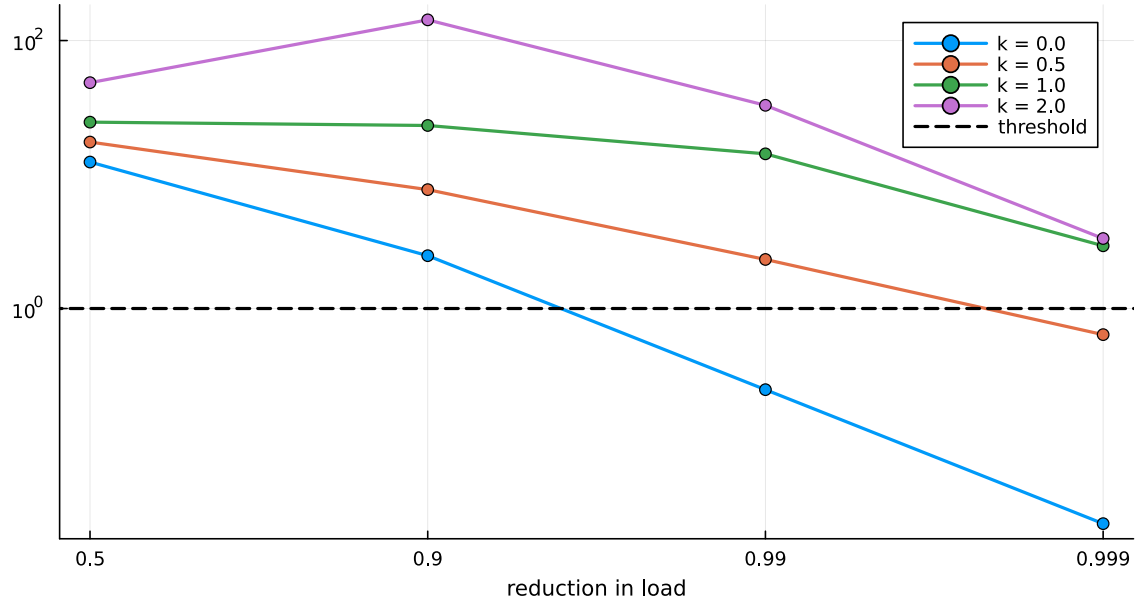
```
Bsal reproduction number if everyone is treated: 232 (untreated 25)
```

```
(step = 6, untreated = 0.039, treated = 0.025)
```

Step 7. Lower load: less death and less shedding

Bacteria that suppress the fungus should lower the load on the animal, and so reduce shedding as well as death. Both now follow from one reduction in load: shedding in proportion to load, death in proportion to load raised to a power k.

```
reductions = [0.5, 0.9, 0.99, 0.999]
ks = [0.0, 0.5, 1.0, 2.0]
R = [effective_reproduction_number(p, load_reducing_treatment(ρ;
 ↪  death_exponent = k), 1.0)
     for ρ in reductions, k in ks]
plot(string.(reductions), R; yscale = :log10, marker = :circle, xlabel =
 ↪  "reduction in load",
    ylabel = "Bsal reproduction number, all treated",
    label = permutedims(["k = $k" for k in ks]), legend = :topright)
hline!([1.0]; color = :black, linestyle = :dash, label = "threshold")
```



With death proportional to load (`k = 1`) the two effects nearly cancel: the animal sheds less but for longer.

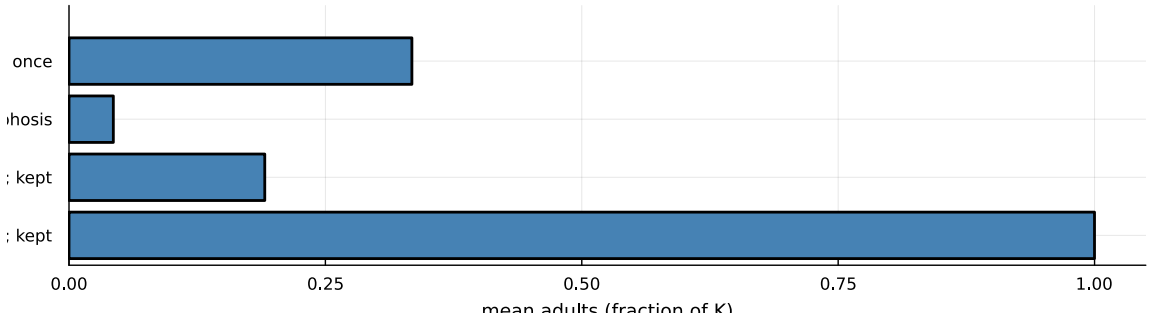
```
a = lar(stage, load_reducing_treatment(0.99; half_life = 1.0, direct =
 ↪  50.0); coverage = 1.0)
step!("Bsal", "treatment", "+ lower load (99%): also less shedding",
 ↪  u3.mean_adults, a.mean_adults,
    "partial; survivors remain infected")
```

```
(step = 7, untreated = 0.039, treated = 0.189)
```

Step 8. Treat larvae as well

With larvae in the host model they can be treated too. Whether a treated larva keeps the bacteria through metamorphosis (`retention`) has not been measured. This step goes back to bacteria that block infection and last, with no spread, to isolate the effect of the stage.

```
quarterly = 0:0.25:9.75
options = [("adults, once", (;)),
    ("larvae, quarterly; lost at metamorphosis", (treat = :larvae,
 ↪  treat_times = quarterly)),
    ("larvae, quarterly; kept", (treat = :larvae, treat_times = quarterly,
 ↪  retention = 1.0)),
    ("both stages, quarterly; kept", (treat = :both, treat_times =
 ↪  quarterly, retention = 1.0))]
vals = [lar(stage, block; coverage = 1.0, kw...).mean_adults for (_, kw) in
 ↪  options]
bar(vals; orientation = :h, yticks = (1:4, first.(options)), yflip = true,
 ↪  legend = false,
    xlabel = "mean adults (fraction of K)", xlims = (0, 1.05), xticks =
 ↪  0:0.25:1,
    left_margin = 6Plots.mm, size = (820, 230), color = :steelblue)
```




```
step!("Bsal", "treatment", "treat both stages, kept through metamorphosis",
 ↪  u3.mean_adults, vals[4],
    "full protection with no spread at all")
```

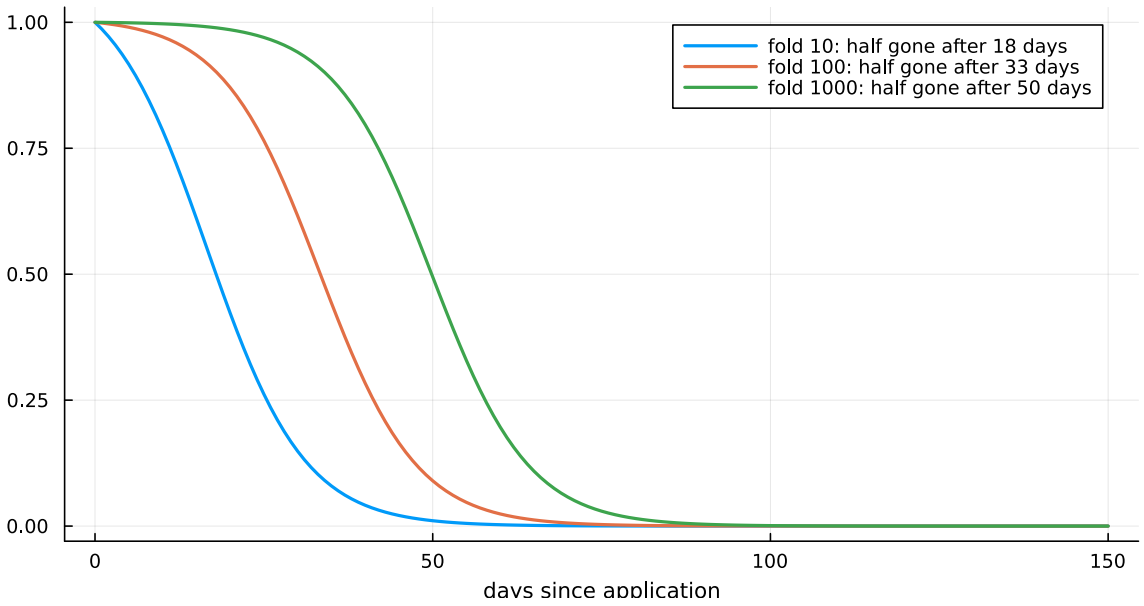
```
(step = 8, untreated = 0.039, treated = 1.0)
```

This overturns step 5. Once recruits can be treated before they become adults, the bacteria do not need to spread.

Step 9. How long the bacteria really last

Two open datasets (a field trial in *Rana sierrae*, Knapp et al. 2022, and a laboratory trial in *Xenopus laevis*, Madison et al. 2025) give the abundance of applied bacteria over time. Fitting an exponential decline gives half-lives of 3 to 14 days, and 4 to 5 days in the field. (The fits are this project's own.) Protection now wanes with abundance. `protective_fold` is how far abundance can fall before half the effect is lost; it has not been measured.

```
days = 0:1:150
plot(xlabel = "days since application", ylabel = "fraction of full
 ↪  protection", legend = :topright)
for F in (10.0, 100.0, 1000.0)
    w = T(efficacy = 0.99, abundance_half_life = 5 / 365, protective_fold =
 ↪  F)
    plot!(days, [protection(w, d / 365) for d in days];
        label = @sprintf("fold %g: half gone after %.0f days", F, 365 *
 ↪  protection_duration(w)))
end
current()
```




```
waning = T(efficacy = 0.99, abundance_half_life = 5 / 365, protective_fold =
 ↪  100.0)
a = lar(stage, waning; coverage = 1.0, treat = :both, treat_times =
 ↪  quarterly, retention = 1.0)
monthly = lar(stage, waning; coverage = 1.0, treat = :both, treat_times =
 ↪  0:(1 / 12):9.95, retention = 1.0)
@printf("dosed monthly instead of quarterly: %.2f\n", monthly.mean_adults)
```

```
step!("Bsal", "treatment", "+ fitted decay (half-life 5 days)",
 ↪  u3.mean_adults, a.mean_adults,
    "fails: protection lasts about a month")
```

```
dosed monthly instead of quarterly: 0.28
```

```
(step = 9, untreated = 0.039, treated = 0.064)
```

Step 10. Chance

Every model so far treats animals as continuous densities, so a population, or Bsal, can be reduced to a fraction of an animal and grow back. The last step runs the same model with whole animals and random events (`StochasticNestedProblem`, which reproduces the equations when the population is large). Bsal arrives as one infected adult in 20 hectares, 500 adults, and each case is run 100 times.

In the stochastic version each class holds a whole number of animals in an area of 20 ha: 500 adults ($K \times 20$) and 64 larvae ($bK/m \times 20$) at the start. Every term of the equations above becomes a per-animal rate, evaluated at densities (count divided by area). In each step of six hours ($\Delta t = 1/1460$ yr) an animal leaves its class with probability $1 - e^{-R\Delta t}$, where R is the sum of its rates out, and goes to each destination with probability in proportion to that rate. Births are Poisson, with mean the birth term of the equations times the area times Δt. A treatment treats each untreated animal with probability ϕ. The transitions come from the package's nested model (`NestedChytridProblem`); with the latent class, larvae and larval infection switched on, as here, they are the equations above, so the two versions agree when the area is large. One difference from the equations: one infected adult in 20 ha is 0.05 per ha, against 0.01 per ha in steps 0–9.

```
using Random: Xoshiro
host_a = ChytridBiology(:wang2021; latent = true, larvae = true,
 ↪  larval_infection = true)
chanceA(tr; kw...) = [solve(StochasticNestedProblem(p, host_a, tr; area =
 ↪  20.0, kw...); rng = Xoshiro(s))
    for s in 1:100]
meanA(rs) = sum(r.mean_adults for r in rs) / length(rs)
cu = chanceA(T())
ct = chanceA(waning; coverage = 1.0, treat = :both, treat_times = quarterly,
 ↪  retention = 1.0)
cb = chanceA(block; coverage = 1.0)
@printf("Bsal lost within 10 years, untreated: %d of 100 runs\n", count(r
 ↪  -> !isnan(r.pathogen_extinction), cu))
@printf("reference treatment of step 3 (block 99%%, adults once): %.2f with
 ↪  chance, %.2f in the equations\n",
    meanA(cb), t3.mean_adults)
step!("Bsal", "chance", "+ chance: whole animals, 20 ha", meanA(cu),
 ↪  meanA(ct),
```

```
    "the failed treatment of step 9 now keeps most of the population")
```

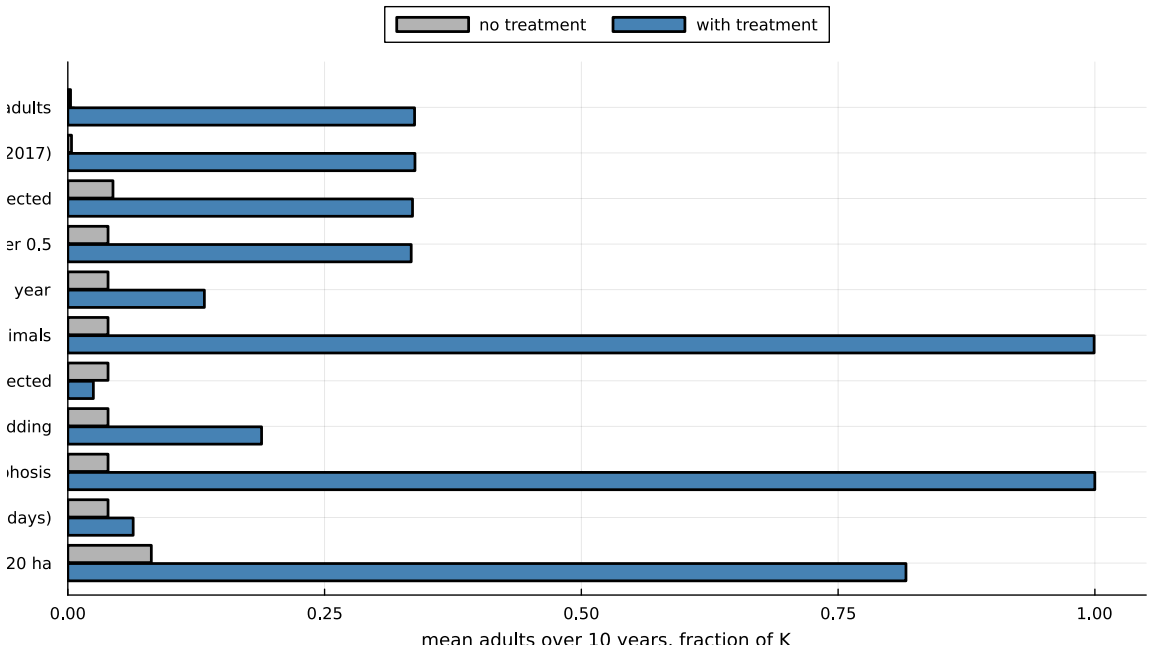
```
Bsal lost within 10 years, untreated: 100 of 100 runs
reference treatment of step 3 (block 99%, adults once): 0.95 with chance, 0.33 in the equat

(step = 10, untreated = 0.081, treated = 0.816)
```

Chance overturns step 9. In the equations, bacteria that last days fail because Bsal never quite disappears: whenever protection lapses it returns from a trace. With whole animals, Bsal arriving in a population that has just been treated usually fails to establish and is lost, and does not come back. The same happens to the step-3 reference treatment, which keeps nearly the whole population instead of a third. The second vignette explores this in detail.

Part A summary

```
ladder_plot("Bsal", "mean adults over 10 years, fraction of K")
```



Part B. Bd in common toads

The outcome in Part B is the number of adults in years 25 to 35 after Bd arrives, as a fraction of the number without Bd. The reference treatment is: 80% of tadpoles treated at hatching every year with bacteria that block 90% of infections for the season. The parameter set is the one behind Figure 2d of Mitchell et al. (2008).

Host and pathogen

Step 0. The pond model

Mitchell's model is the simplest available for this system, though it is not as simple as step 0 of Part A: tadpoles are infected by zoospores in the pond, infected tadpoles are removed at metamorphosis, and juveniles and adults are never infected.

The equations below are the host model as it stands after step 3. Time t is in years and T is the time of year. Between the yearly events,

$$\begin{aligned}
\frac{dP_X}{dt} &= -(\mu_P + \nu Z)\, P_X \\
\frac{dP_Y}{dt} &= \nu Z\, P_X - (\mu_P + \alpha_P)\, P_Y \\
\frac{dM}{dt} &= -(\mu_J + \varepsilon)\, M \\
\frac{dA_X}{dt} &= \varepsilon M \left(1 - \frac{A_X + A_Y}{K}\right) - \mu_A A_X - \chi(t)\, \nu_A Z A_X + \sigma_A A_Y \\
\frac{dA_Y}{dt} &= \chi(t)\, \nu_A Z A_X - (\mu_A + \alpha_A + \sigma_A) A_Y \\
\frac{dZ}{dt} &= \rho P_Y + \chi(t)\, \rho_A A_Y + \gamma Z e^{-\varphi Z} - \mu_Z Z
\end{aligned}$$

where $\chi(t) = 1$ while adults are at the pond ($t_1 \le T < t_1 + d_A/365$) and 0 otherwise. Three events happen each year:

$$\begin{aligned}
T = t_1 &: \quad P_X \leftarrow P_X + \iota\psi\,(A_X + A_Y) \\
T = t_2 &: \quad J_0 = P_X\,(1 - q + q\,s_J) + P_Y\,s_J, \qquad q = 1 - e^{-\nu Z d_J/365}, \qquad P_X,\ P_Y \leftarrow 0 \\
\omega \text{ after } t_2 &: \quad M \leftarrow M + J_0\, e^{-\mu_J \omega}
\end{aligned}$$

At breeding, tadpoles are born to every adult. At metamorphosis the tadpoles leave the pond as a juvenile cohort J_0; Z in q is its value at that moment. A cohort only dies (at rate μ_J) until it is ω old, and then joins the pool M of juveniles that can mature into adults. M is Mitchell's $\int_\omega^\infty J\,da$: following each cohort in closed form solves the paper's age-structured juvenile equation exactly.

Step	What it adds	Terms
0	Mitchell et al. (2008), equations (2.1) to (2.5)	$s_J = 0$, $d_J = 0$, no A_Y ($\chi = 0$)
1	infected metamorphs can survive	$s_J > 0$
2	metamorphs are exposed as they leave	$d_J > 0$, so $q > 0$

Step	What it adds	Terms
3	adults are infected at the pond, shed there and clear the infection	A_Y, $\chi(t)$, ν_A, ρ_A, σ_A

With steps 1 to 3 switched off, the equations are those of Mitchell et al. (2008), and the package reproduces the outcomes of their Fig. 2b to 2d. Bd arrives as Z_0 zoospores at the start of year 0, after 100 years without Bd. Following the paper, the toads are declared extinct (every stage set to zero) below 50 adults, and Bd below 100 zoospores. The package tests the toads at each yearly event, and Bd only at breeding, when there are no tadpoles. In step 3 Bd is not removed at breeding while an infected adult is present.

Symbol	Meaning	Value or source
P_X, P_Y	uninfected and infected tadpoles	
J_0, M	a juvenile cohort at metamorphosis; juveniles older than ω	
A_X, A_Y	uninfected and infected adults	
Z	free-living zoospores in the pond	
t_1, t_2	day of breeding and of metamorphosis	days 60 and 155, compendium (mitchell2008)
ω	age at which juveniles can start to mature	500 days, compendium (mitchell2008)
ι, ψ	mating couples per adult; eggs per couple	0.14, 2000, compendium (mitchell2008)
ε	maturation rate of juveniles older than ω	0.16 per year, compendium (mitchell2008)
μ_P, μ_J, μ_A	natural death rates of tadpoles, juveniles, adults	7.55, 0.73, 0.73 per year, compendium (mitchell2008)
K	limit on the number of adults	14,000, compendium (mitchell2008)
α_P	extra death rate of infected tadpoles	1.62 per year, compendium (mitchell2008)
ν	infection rate per zoospore	2.3×10^{-8} per zoospore per year, compendium (mitchell2008, Fig. 2d)
ρ	zoospores released per infected tadpole	6×10^6 per tadpole per year, compendium (mitchell2008, Fig. 2d)
μ_Z	zoospore death rate	13 per year, compendium (mitchell2008)

Symbol	Meaning	Value or source
γ, φ	saprobic growth of zoospores and its density dependence	0 (so the term is off), 10^{-6}, compendium (mitchell2008)
Z_0	zoospores introduced	10^4, package default
s_J	chance that an infected metamorph survives and clears Bd	0 (step 0); $1 - 72/101 = 0.29$ from step 1, compendium (smith2022)
d_J	days metamorphs are exposed as they leave	0 (steps 0, 1); 7 from step 2, illustrative
d_A	days adults spend at the pond from t_1	30 in step 3, illustrative
ν_A, ρ_A	adult infection rate and shedding while at the pond	equal to ν and ρ in step 3, illustrative
σ_A	adult clearance rate	1 per year in step 3, illustrative
α_A	extra death rate of infected adults	0, illustrative

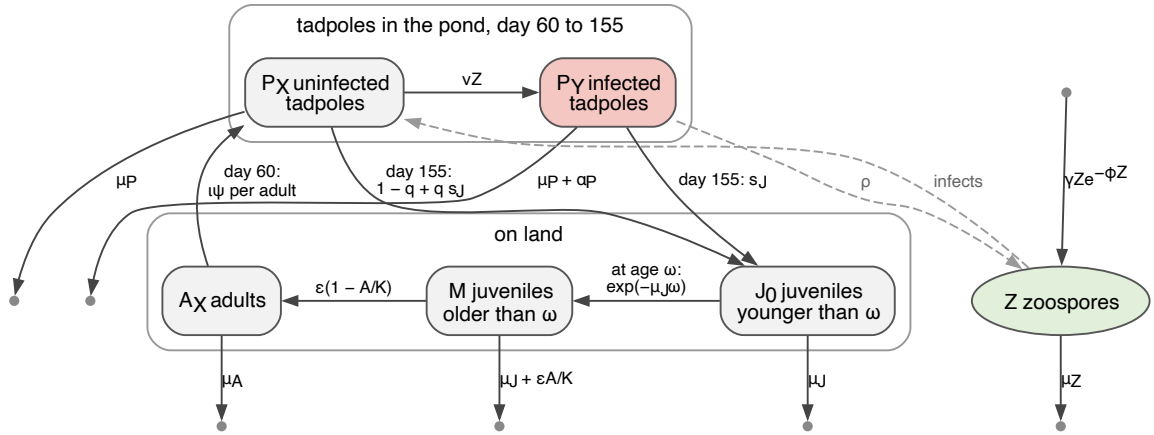


Figure 4: Steps 0 to 2: the life cycle and infection of tadpoles. Mitchell et al. (2008) is the case $s_J = 0$ and $q = 0$. Breeding (day 60), metamorphosis (day 155) and maturation at age ω are yearly pulses; dashed arrows are shedding into and infection from the zoospore pool. $A = A_X + A_Y$.

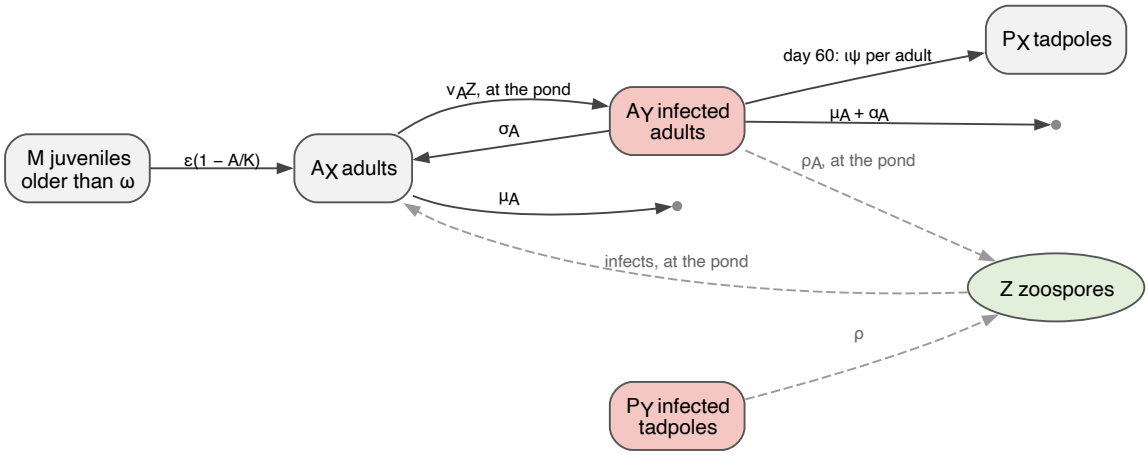


Figure 5: Step 3: adults are infected, shed and clear Bd while at the pond, and infected adults breed like uninfected ones.

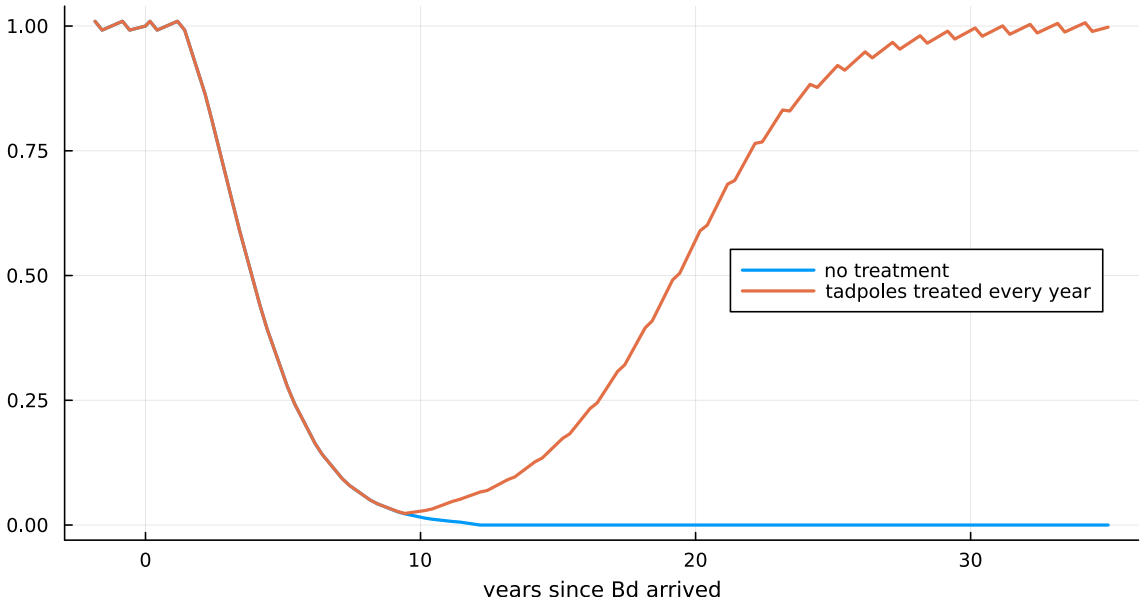
```
m = MitchellParameters(:mitchell2008; set = "fig2d")
none = PostMetamorphicInfection()
pond(tr; stages = none, kw...) = solve(ProbioticPondProblem(m, tr; stages,
 ↪  kw...))
rel(r) = r.A_mean / r.A_disease_free
ref = T(efficacy = 0.9)

u, t = pond(T()), pond(ref; coverage = 0.8)
@printf("Bd persists: untreated %s, treated %s\n", u.bd_persist,
 ↪  t.bd_persist)
step!("Bd toad", "host", "Mitchell 2008 pond model", rel(u), rel(t),
    "untreated: toads and Bd both die out; treated: toads saved, Bd gone")
```

```
Bd persists: untreated false, treated false
```

```
(step = 0, untreated = 0.0, treated = 0.975)
```

```
series(r) = (r.t[r.t .>= -2], r.A[r.t .>= -2] ./ r.A_disease_free)
plot(series(u)...; label = "no treatment", xlabel = "years since Bd
 ↪  arrived",
    ylabel = "adults (fraction of disease-free)", legend = :right)
plot!(series(t)...; label = "tadpoles treated every year")
```



Unlike Bsal, the simplest treatment works here: the infected stage is short and a new cohort can be treated each year. It does not prevent the initial crash.

Step 1. Infected metamorphs can survive

Experiments show that toadlets exposed to Bd are infected, that most die, and that some survive (Smith et al. 2022; counts from the compendium).

```
s = parameter_set(:smith2022)
survival = 1 - s.deaths_exposed / s.n_exposed
@printf("toadlet survival after exposure: %.2f\n", survival)
st1 = PostMetamorphicInfection(metamorph_survival = survival)
u, t = pond(T(); stages = st1), pond(ref; coverage = 0.8, stages = st1)
@printf("Bd persists: untreated %s, treated %s\n", u.bd_persist,
 ↪  t.bd_persist)
step!("Bd toad", "host", "+ infected metamorphs can survive (0.29)", rel(u),
 ↪  rel(t),
    "toads no longer die out, so nor does Bd; treatment now protects
     ↪  poorly")
```

```
toadlet survival after exposure: 0.29
Bd persists: untreated true, treated true
```

```
(step = 1, untreated = 0.046, treated = 0.085)
```

One added survival probability reverses both outcomes. The published “both die out” result depended on every infected animal dying; once some survive, the host persists, Bd persists with it, and blocking 90% of infections is no longer enough.

Step 2. Metamorphs are exposed as they leave the pond

```
st2 = PostMetamorphicInfection(metamorph_survival = survival,
 ↪  metamorph_exposure_days = 7.0)
u, t = pond(T(); stages = st2), pond(ref; coverage = 0.8, stages = st2)
step!("Bd toad", "host", "+ metamorphs exposed for a week on leaving",
 ↪  rel(u), rel(t),
    "little further change")
```

```
(step = 2, untreated = 0.046, treated = 0.082)
```

Step 3. Adults carry Bd between years

Breeding adults carry Bd at low prevalence in field surveys. Here adults are infected while at the pond, shed there, and clear the infection over about a year. (Treatment is not yet combined with infected adults in the package, so only the untreated outcome is shown.)

```
st3 = PostMetamorphicInfection(metamorph_survival = survival,
 ↪  metamorph_exposure_days = 7.0,
    adult_exposure_days = 30.0, adult_relative_shedding = 1.0,
 ↪  adult_clearance = 1.0)
r = solve(ToadStageProblem(m, st3))
@printf("Bd persists: %s; adult prevalence after breeding: %.2f\n",
 ↪  r.bd_persist, r.adult_prevalence)
step!("Bd toad", "host", "+ adults carry Bd between years", r.A_mean /
 ↪  r.A_disease_free, NaN,
    "untreated only; little change in this parameter set")
```

```
Bd persists: true; adult prevalence after breeding: 0.99
```

```
(step = 3, untreated = 0.044, treated = NaN)
```

The host model of step 2 is used for the rest of Part B.

Every "with treatment" outcome in Part B, including the reference treatment of steps 0 to 2, comes from the model below. From step 4 it sits on the step-2 host, so there are no infected adults ($A_Y = 0$). Tadpoles are split into untreated (U) and treated (T), uninfected and infected:

$$
\begin{aligned}
\frac{dP_{XU}}{dt} &= -(\mu_P + \nu Z)\, P_{XU} + \lambda P_{XT} \\
\frac{dP_{XT}}{dt} &= -\big(\mu_P + (1-g\eta)\,\nu Z + \lambda\big)\, P_{XT} \\
\frac{dP_{YU}}{dt} &= \nu Z\, P_{XU} - (\mu_P + \alpha_P)\, P_{YU} + \lambda P_{YT} \\
\frac{dP_{YT}}{dt} &= (1-g\eta)\,\nu Z\, P_{XT} - \big(\mu_P + (1-g\tau)\,\alpha_P + \lambda\big)\, P_{YT} \\
\frac{dZ}{dt} &= \rho\big(P_{YU} + (1-g\kappa)\, P_{YT}\big) + \gamma Z e^{-\varphi Z} - \mu_Z Z
\end{aligned}
$$

with M and A_X as before. A treated tadpole is infected less often (efficacy η), dies of infection less often (tolerance τ) and sheds less (shedding reduction κ). It loses the bacteria at rate $\lambda = \ln 2/h$. All three effects are scaled by the protection left since the last dose, $g = g(t - t_{\text{dose}})$, where

$$
g(a) = \frac{1 + 1/F}{1 + 2^{a/h_B}/F}.
$$

$g = 1$ when the bacteria do not decline ($h_B = \infty$). Treatment enters through three events in each year from the arrival of Bd:

$$
\begin{aligned}
T = t_1 :&\quad P_{XU} \leftarrow P_{XU} + \iota\psi A_X, \quad \text{then} \quad P_{XT} \leftarrow P_{XT} + c\,P_{XU},\ P_{XU} \leftarrow (1-c)\,P_{XU},\ t_{\text{dose}} \leftarrow t \\
T = t_1 + d_k/365 :&\quad P_{iT} \leftarrow P_{iT} + c\,P_{iU},\ P_{iU} \leftarrow (1-c)\,P_{iU}\ \ (i = X, Y),\ t_{\text{dose}} \leftarrow t \\
T = t_2 :&\quad J_0 = P_{XU}(1 - q_U + q_U s_U) + P_{XT}(1 - q_T + q_T s_T) + P_{YU} s_U + P_{YT} s_T
\end{aligned}
$$

with $q_U = 1 - e^{-\nu Z d_J/365}$, $q_T = 1 - e^{-(1-g\eta)\nu Z d_J/365}$, $s_U = s_J$ and $s_T = s_J + g\tau(1 - s_J)$. Tolerance therefore also raises the survival of infected metamorphs. Metamorphs lose the bacteria, so every juvenile and adult is untreated. With one dose at hatching and $h_B = 5$ days, $F = 100$, the protection left at metamorphosis 95 days later is $g = 1.01/(1 + 2^{19}/100) \approx 0.0002$. Each extra dose also treats a share c of the tadpoles still untreated, so after the dose at hatching and four more, all but 0.2^5 (about 0.03%) of tadpoles have been treated.

The steps do not all add to the one before: steps 5 to 7 change which effect the bacteria have.

Step	Treatment in the "with treatment" outcome
0 to 2 (reference)	$\eta = 0.9$, $c = 0.8$, no loss of protection
3	none (treatment is not yet combined with infected adults)
4	$\eta = 0.9$, $h = 0.05$ years (18 days)
5	$\tau = 0.7$ in place of η; no loss of protection
6	$\tau = \kappa = 0.7$ (`load_reducing_treatment(0.7)`)
7	$\tau = 0.7$, declining with $h_B = 5$ days and $F = 100$; one dose

Step	Treatment in the "with treatment" outcome
8	as step 7, with doses also on days $d_k = 20$, 40, 60 and 80
9	as step 8, with chance

Symbol	Meaning	Value or source
c	share of untreated tadpoles treated at each dose	0.8, illustrative
η	efficacy: reduction in the infection rate	0.9 (reference, step 4); 0.5, 0.9, 0.99 compared in step 5; illustrative
τ	tolerance: reduction in the extra death rate of infected tadpoles	0.7 (steps 5 to 9); 0.3, 0.7, 1.0 compared in step 5; illustrative
κ	shedding reduction	0.7 in step 6 (load 70% lower, death proportional to load), illustrative
h, λ	half-life of the treated state; $\lambda = \ln 2/h$	18 days in step 4, otherwise ∞; illustrative
h_B	half-life of the applied bacteria	5 days, this project's fit to Knapp et al. (2022) and Madison et al. (2025) (Part A, step 9)
F	starting abundance over the abundance giving half the effect	100, illustrative
d_k	days after hatching of extra doses	20, 40, 60, 80 in step 8, illustrative
g	share of the full effect left since the last dose	from h_B and F

The package can also spread the bacteria between tadpoles, through a pond reservoir, or from treated mothers, and can treat adults instead. All of these are off in Part B.

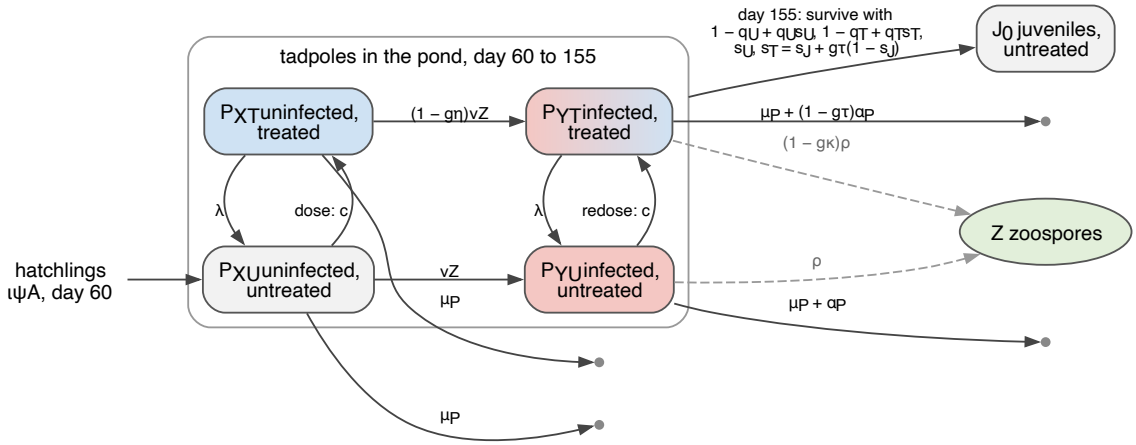


Figure 6: Treated tadpoles (steps 4 to 9). Each dose moves a share c of untreated tadpoles to the treated classes (doses after hatching from step 8); λ is loss of the bacteria (step 4); g is the protection left since the last dose. Juveniles and adults are as in the first diagram, and all are untreated.

Treatment

Step 4. Protection is lost

```
u = pond(T(); stages = st2)
a = pond(T(efficacy = 0.9, half_life = 0.05); coverage = 0.8, stages = st2)
step!("Bd toad", "treatment", "+ protection lost, half-life 18 days",
 ↪  rel(u), rel(a), "slightly worse")
```

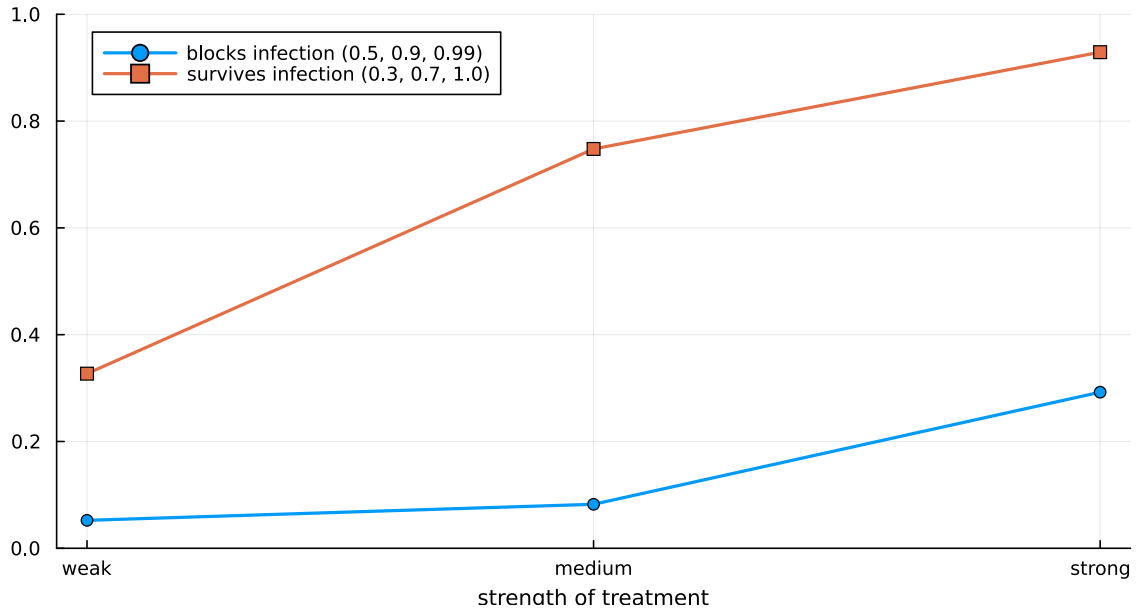
```
(step = 4, untreated = 0.046, treated = 0.056)
```

Step 5. Mechanism as in trials: survival, not fewer infections

This step swaps the mechanism: treated tadpoles are infected as often as untreated ones but survive better. It also sets aside the loss of protection of step 4, so that the two mechanisms are compared on equal terms; steps 5 to 8 keep bacteria for life unless their decay is stated.

```
effs, tols = [0.5, 0.9, 0.99], [0.3, 0.7, 1.0]
blocking = [rel(pond(T(efficacy = e); coverage = 0.8, stages = st2)) for e
 ↪  in effs]
tolerating = [rel(pond(T(tolerance = τ); coverage = 0.8, stages = st2)) for
 ↪  τ in tols]
plot(["weak", "medium", "strong"], blocking; marker = :circle,
    label = "blocks infection (0.5, 0.9, 0.99)", xlabel = "strength of
 ↪  treatment",
```

```
    ylabel = "adults (fraction of disease-free)", ylims = (0, 1), legend =
 ↪  :topleft)
plot!(["weak", "medium", "strong"], tolerating; marker = :square,
    label = "survives infection (0.3, 0.7, 1.0)")
```




```
a = pond(T(tolerance = 0.7); coverage = 0.8, stages = st2)
step!("Bd toad", "treatment", "mechanism as in trials: survive, still
 ↪  infected", rel(u), rel(a),
    "much better than blocking here; Bd persists")
```

```
(step = 5, untreated = 0.046, treated = 0.748)
```

For Bsal the observed mechanism made treatment fail. Here it makes treatment work better, because Bd is going to persist anyway and what matters is how many infected animals live.

Step 6. Lower load

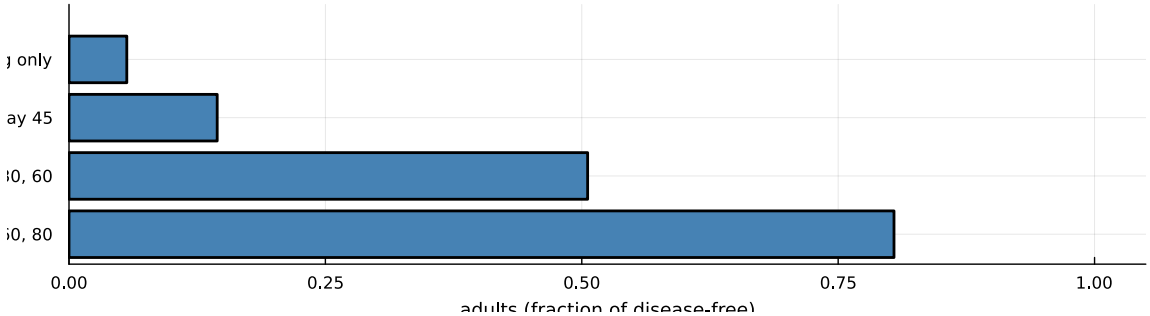
```
a = pond(load_reducing_treatment(0.7); coverage = 0.8, stages = st2)
step!("Bd toad", "treatment", "+ lower load (70%): also less shedding",
 ↪  rel(u), rel(a),
    "little changed: untreated tadpoles and the pond keep Bd going")
```

```
(step = 6, untreated = 0.046, treated = 0.751)
```

Step 7. How long the bacteria really last

With the fitted decay (half-life 5 days), one dose at hatching wanes during the 95-day tadpole season. This step keeps the survival mechanism of step 5 but not the lower shedding of step 6.

```
waning_t = T(tolerance = 0.7, abundance_half_life = 5 / 365, protective_fold
 ↪  = 100.0)
schedules = [("at hatching only", Float64[]), ("+ day 45", [45.0]), ("+ days
 ↪  30, 60", [30.0, 60.0]),
    ("+ days 20, 40, 60, 80", [20.0, 40.0, 60.0, 80.0])]
doses = [rel(pond(waning_t; coverage = 0.8, stages = st2, redose_days = d))
 ↪  for (_, d) in schedules]
bar(doses; orientation = :h, yticks = (1:4, first.(schedules)), yflip =
 ↪  true, legend = false,
    xlabel = "adults (fraction of disease-free)", xlims = (0, 1.05), xticks
 ↪  = 0:0.25:1,
    left_margin = 6Plots.mm, size = (820, 230), color = :steelblue)
```




```
step!("Bd toad", "treatment", "+ fitted decay, one dose at hatching",
 ↪  rel(u), doses[1],
    "protection gone before the season ends")
```

```
(step = 7, untreated = 0.046, treated = 0.056)
```

Step 8. Several doses in a season

```
step!("Bd toad", "treatment", "+ five doses per season", rel(u), doses[4],
 ↪  "most of the benefit restored")
```

```
(step = 8, untreated = 0.046, treated = 0.804)
```

Each dose renews the protection of treated tadpoles and also treats 80% of those still untreated, infected ones included, so after five doses almost every tadpole has been treated. Part of the benefit comes from that higher coverage, not only from renewed protection.

Step 9. Chance

The same pond model with every tadpole, toad and zoospore a whole number, and every event random (`solve(problem; rng)`); 50 runs of each case. Mitchell et al.'s rule that toads are extinct below 50 adults, and Bd below 100 zoospores, is no longer needed: each goes extinct when its numbers reach zero.

In terms of the equations, every flow above becomes a random event, drawn in steps of one day (tau-leaping). In each step every animal or zoospore leaves its class with probability $1 - e^{-Rh}$, where R is the sum of its per-capita rates out of that class and h is one day, and goes to each destination in proportion to that rate. New zoospores are a Poisson draw with mean $\rho\,(P_{YU} + (1 - g\kappa)P_{YT})\,h$. The yearly events become draws as well: hatchlings are Poisson with mean $\iota\psi A_X$, each dose treats a binomial share c of untreated tadpoles, and an uninfected, untreated metamorph reaches age ω with probability $(1 - q_U + q_U s_U)\,e^{-\mu_J \omega}$ (likewise for the other classes). The limits of 50 adults and 100 zoospores are not applied.

```
chanceB(tr; stages = st2, kw...) = [solve(ProbioticPondProblem(m, tr;
 ↪  stages, kw...); rng = Xoshiro(s)) for s in 1:50]
meanB(rs) = sum(rel(r) for r in rs) / length(rs)
five = (coverage = 0.8, redose_days = [20.0, 40.0, 60.0, 80.0])
bu, bt = chanceB(T()), chanceB(waning_t; five...)
base = chanceB(T(); stages = none)
blk = chanceB(ref; coverage = 0.8)
@printf("Mitchell's model (step 0), untreated: toads persist in %d of 50
 ↪  runs, adults %.2f (equations: extinct)\n",
    count(r -> r.toads_persist, base), meanB(base))
@printf("reference treatment (block 90%%) on the step-2 host: %.2f, Bd
 ↪  persists in %d of 50 (equations: %.2f)\n",
    meanB(blk), count(r -> r.bd_persist, blk), rel(pond(ref; coverage = 0.8,
     ↪  stages = st2)))
step!("Bd toad", "chance", "+ chance: whole animals", meanB(bu), meanB(bt),
    "little change for this model; large changes elsewhere (see text)")
```

```
Mitchell's model (step 0), untreated: toads persist in 50 of 50 runs, adults 0.78 (equation
reference treatment (block 90%) on the step-2 host: 0.36, Bd persists in 35 of 50 (equation
```

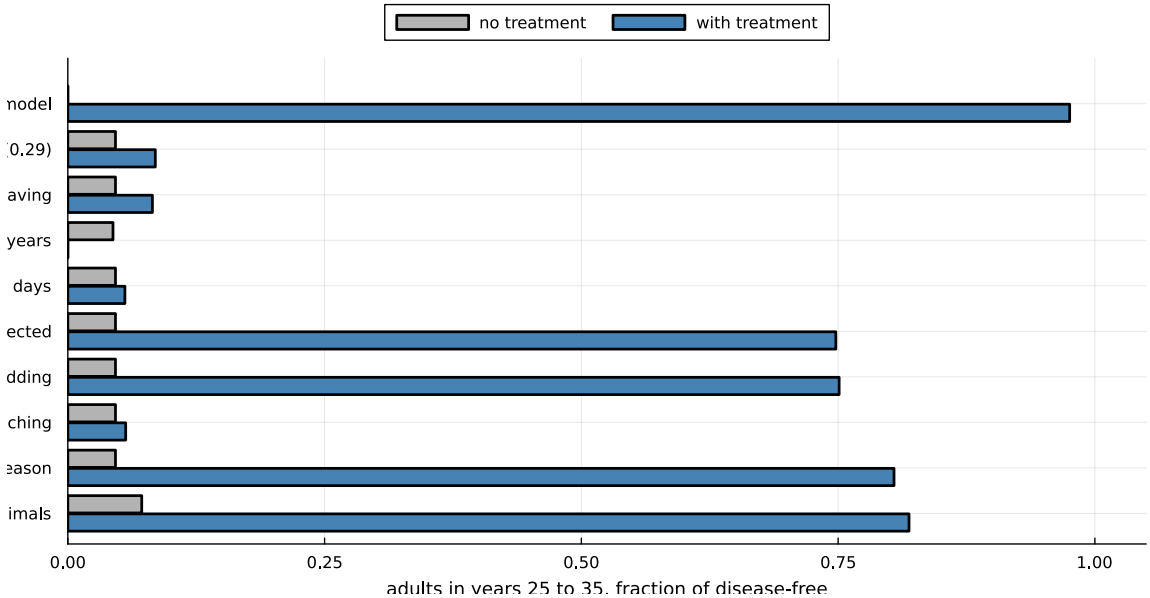
```
(step = 9, untreated = 0.072, treated = 0.819)
```

For the final model chance changes little. Elsewhere on the ladder it changes a great deal:

- Mitchell's published outcome does not survive it. In the equations Bd drives the toads to extinction and then dies out itself. With whole animals the toads fall to a few dozen adults, too few tadpoles remain to carry Bd through the winter, Bd is lost, and the toads recover. The published result depends on declaring toads extinct below 50 adults.
- Blocking infection does better. On the step-2 host model, Bd is lost by chance in some runs, and treatment then protects the population fully.

Part B summary

```
ladder_plot("Bd toad", "adults in years 25 to 35, fraction of disease-free")
```



Part C. Bd in frogs

The outcome in Part C is the fraction of 80 frogs sharing one pool that are alive after two years. Disease here depends on the fungal load on each frog. The reference treatment is introduced at step 2.

Host and pathogen

Step 0. Load on each frog

In the model of Briggs et al. (2010) each frog carries a load of sporangia. Sporangia release zoospores into a shared pool, zoospores infect frogs, and a frog dies when its load reaches a lethal level. Loads run away when there are enough frogs to keep the pool topped up.

The equations below are the model as it stands after step 2. Each live frog i carries S_i sporangia, and the pool holds Z zoospores:

$$\frac{dS_i}{dt} = \frac{\nu\gamma}{V} Z + (\eta\nu f - \sigma) S_i - \frac{c_i S_i}{h + S_i},$$
$$\frac{dZ}{dt} = \varepsilon + \eta(1-f) \sum_{j \text{ alive}} S_j - \mu Z - \frac{\gamma}{V} N_{\text{alive}} Z.$$

Zoospores meet each frog at rate γ/V, and a fraction ν of meetings start a sporangium. Each sporangium releases η zoospores a day: a fraction f reinfect the same frog and the rest go to the pool. A frog dies when S_i reaches $S_{\max}$; from then on it sheds nothing and takes up no zoospores. All frogs start with no sporangia, and the pool starts at $\varepsilon/(\mu + \gamma N/V)$. On its own a frog's load would decline, because $\eta\nu f$ (about 0.04 per day) is below σ; loads run away only when enough frogs feed the pool.

What each step adds:

- Step 0 is the model of Briggs et al. (2010) with no clearance ($c_i = 0$). The S_i and Z equations without ε are those of the paper. The outside input ε is the paper's external source of zoospores (its Fig. 3D), here as a constant rate.
- Step 1 adds clearance by the frog, $c\,S_i/(h + S_i)$, the same for every frog ($c_i = c$). This term is new in this package.
- Step 2 lets c_i differ: the 80 values are evenly spaced quantiles of a lognormal distribution with median c.

Symbol	Meaning	Value or source
S_i	sporangia on frog i	state (sporangia per frog)
Z	zoospores in the pool	state (zoospores per pool)
N_{alive}	frogs alive	starts at $N = 80$
V	pool volume	1 volume unit; compendium (briggs2010)
γ	rate at which zoospores meet a frog	0.01 volume units per day; compendium (briggs2010)
ν	fraction of meetings that start a sporangium	0.05; compendium (briggs2010)
f	fraction of released zoospores that reinfect the same frog	0.05; compendium (briggs2010)
η	zoospores released per sporangium	17.5 per day; compendium (briggs2010)
σ	loss of sporangia	0.2 per day; compendium (briggs2010)
μ	death of zoospores in the pool	1 per day; compendium (briggs2010)
$S_{\max}$	lethal load	10,000 sporangia; compendium (briggs2010)
ε	zoospores entering from outside	1,000 per day; set in this vignette (the value Briggs et al. used for an external source)
c	maximum clearance (step 1)	100 sporangia per day; illustrative
h	load at which clearance is half its maximum (step 1)	500 sporangia; illustrative

Symbol	Meaning	Value or source
c_i	clearance of frog i (step 2)	lognormal, median c, log-scale sd 0.5; illustrative

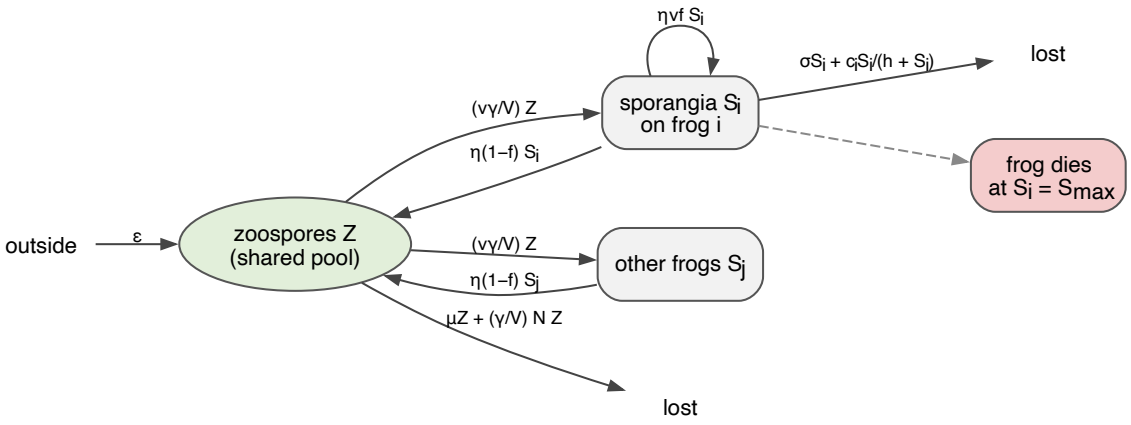


Figure 7: The load model for one frog i and the shared zoospore pool. Clearance (c_i) is added in steps 1 and 2.

```
pb = BriggsParameters(:briggs2010)
N, hsat = 80, 500.0
ε = parameter_set(:briggs2010; set = "fig3d").εZ    # zoospores from
 ↪  outside, per day
tipping(c; n = N) = solve(LoadTippingProblem(pb; c, h = hsat, N = n, ε,
 ↪  tspan = (0.0, 730.0)))
alive(r) = count(r.alive) / length(r.alive)

@printf("frogs needed for loads to run away: %.0f\n",
 ↪  host_density_threshold(pb))
r0 = tipping(0.0)
step!("Bd frog", "host", "Briggs 2010 load model", alive(r0), NaN, "all die:
 ↪  80 frogs is above the threshold of 23")
```

```
frogs needed for loads to run away: 23

(step = 0, untreated = 0.0, treated = NaN)
```

Step 1. Frogs clear some of their load

Frogs have defences. Here each frog removes sporangia at a rate that saturates as load rises: defences cope with a light infection and are overwhelmed by a heavy one (this package's load tipping model; the clearance values are illustrative).

```
c = 100.0
@printf("frogs needed for loads to run away: %d\n", critical_host_number(pb;
 ↪  c, h = hsat, ε))
r1 = tipping(c)
r1small = tipping(c; n = 50)
@printf("alive after two years: %.2f of 80 frogs, %.2f of 50 frogs\n",
 ↪  alive(r1), alive(r1small))
step!("Bd frog", "host", "+ saturating clearance by the frog", alive(r1),
 ↪  NaN,
    "threshold rises from 23 to 68 frogs; 80 still all die")
```

```
frogs needed for loads to run away: 68
alive after two years: 0.00 of 80 frogs, 1.00 of 50 frogs
```

```
(step = 1, untreated = 0.0, treated = NaN)
```

Step 2. Frogs differ

Frogs now differ in how strong their defences are (lognormal, same median). The reference treatment for Part C starts here: each frog, with probability 0.4, carries bacteria that triple its clearance, for life.

```
cs = quantile.(LogNormal(log(c), 0.5), ((1:N) .- 0.5) ./ N)
r2 = tipping(cs)
frogs(tr; kw...) = solve(ProbioticLoadProblem(pb, tr; c, c_treated = 3c, h
 ↪  = hsat, N, ε, days = 730, kw...);
    rng = Xoshiro(1))
ref_c = frogs(T(); coverage = 0.4)
step!("Bd frog", "host", "+ frogs differ in their defences", alive(r2), NaN,
    "a few of the best-defended survive")
```

```
(step = 2, untreated = 0.062, treated = NaN)
```

With identical frogs a die-off is all or nothing; with differences it is partial, and it stops when too few frogs are left to sustain the pool. The treatment steps use identical frogs (step 1), because treatment and host differences are not yet combined in the package.

Treatment

Step 3. A treated class

Treated frogs clear sporangia faster. In the load equations above, with identical frogs,

$$c_i = \begin{cases} c_T, & \text{frog } i \text{ treated,} \\ c, & \text{otherwise.} \end{cases}$$

On day 0 each frog is treated with probability equal to the coverage, so the treated frogs differ from run to run. Loads follow the equations through each day. At the end of each day a treated frog stops being treated with probability $1 - e^{-\omega}$, where $\omega = \ln 2 / t_{1/2}$. In step 3, $t_{1/2}$ is infinite (treated for life); step 4 sets $t_{1/2} = 30$ days and treats every frog. The package can also let untreated frogs pick up the bacteria from treated ones; that is switched off here.

Symbol	Meaning	Value or source
c_T	clearance of a treated frog	$3c = 300$ sporangia per day; illustrative
coverage	probability that a frog is treated on day 0	0.4 (step 3), 1 (step 4); illustrative
$t_{1/2}$	half-life of the treated state	infinite (step 3), 30 days (step 4); illustrative

```
step!("Bd frog", "treatment", "treated class: 40% of frogs, lasts for life",
 ↪  alive(r1),
    ref_c.alive[end] / N, "all survive, including the untreated: protection
 ↪  is shared")
```

```
(step = 3, untreated = 0.0, treated = 1.0)
```

Because disease depends on the shared pool, treating 40% protects everyone.

Step 4. Protection is lost

```
a = frogs(T(half_life = 30.0); coverage = 1.0)
@printf("first death: day %d (untreated: day %d)\n", minimum(filter(>(0),
 ↪  a.death_day)),
    ceil(Int, minimum(r1.death_time)))
step!("Bd frog", "treatment", "+ protection lost, half-life 30 days",
 ↪  alive(r1), a.alive[end] / N,
    "die-off delayed by about three months, not prevented")
```

```
first death: day 241 (untreated: day 156)
```

```
(step = 4, untreated = 0.0, treated = 0.0)
```

Step 5. Bacteria as a state of each frog

Until now every treated animal has been simply "treated". Each frog now carries its own abundance of the applied bacteria. It declines with the half-life fitted to field data (5 days; this project's own fit), and frogs exchange bacteria through the pond. Exchange moves bacteria between frogs without creating any, so they spread but do not sustain themselves. That is what the field trial showed: eleven days after most frogs in a pond were bathed, frogs that were never bathed carried about 0.63 times as much of the bacterium as bathed ones, and it then declined in both.

```
baths(; x = 0.1, F = 100.0, coverage = 0.75, dose_days = [0], days = 730) =
    solve(ProbioticAbundanceProblem(pb, T(abundance_half_life = 5.0,
     ↪  protective_fold = F, direct = x);
        c, c_treated = 3c, h = hsat, N, ε, coverage, dose_days, days))
for x in (0.0, 0.1, 1.0)
    r = baths(; x, days = 30)
    @printf("exchange %.1f per day: unbathed/bathed abundance on day 11 =
     ↪  %.2f\n", x,
        r.abundance_unbathed[12] / r.abundance_bathed[12])
end
```

```
exchange 0.0 per day: unbathed/bathed abundance on day 11 = 0.00
exchange 0.1 per day: unbathed/bathed abundance on day 11 = 0.60
exchange 1.0 per day: unbathed/bathed abundance on day 11 = 1.00
```

An exchange rate of about 0.1 per day reproduces the field ratio and is used below.

In equations, each frog i now carries an abundance B_i of the applied bacteria, in dose units. A bath sets $B_i = 1$ on every live frog in the bathed group (the first 75% of frogs, 60 of 80). Between baths

$$\frac{dB_i}{dt} = -\delta B_i + x\,(\bar{B} - B_i), \qquad \delta = \frac{\ln 2}{t_B}, \qquad \bar{B} = \frac{1}{N_{\text{alive}}} \sum_{j\ \text{alive}} B_j,$$

$$c_i = c + (c_T - c)\,\frac{B_i}{B_i + 1/F},$$

and S_i and Z follow the equations of step 1 with this c_i. Summed over live frogs the exchange terms cancel, so the mean abundance falls at rate δ whatever x is. With $F = 100$ a freshly bathed frog has 99% of the extra clearance, and keeps at least half of it until B_i falls to $1/F$: about $\log_2 F \approx 6.6$ half-lives (33 days) without exchange. The treated class of steps 3 and 4 is not part

of this model; protection now wanes through B_i. Step 6 repeats the bath on days 0, 90, 180, and so on.

Symbol	Meaning	Value or source
B_i	abundance of applied bacteria on frog i	state (dose units; 1 just after a bath)
t_B	half-life of the abundance	5 days; this package's fit to field data, not a published estimate
x	rate at which frogs exchange bacteria	0.1 per day; chosen to match the field ratio above
F	protective fold: abundance after a bath divided by the abundance giving half the effect	100; illustrative
c_T	clearance at full effect	$3c = 300$ sporangia per day; illustrative
coverage	share of frogs bathed	0.75; illustrative

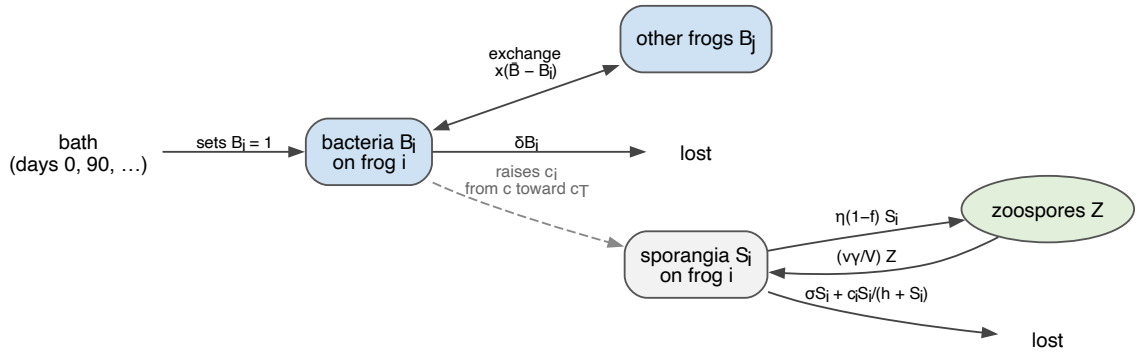


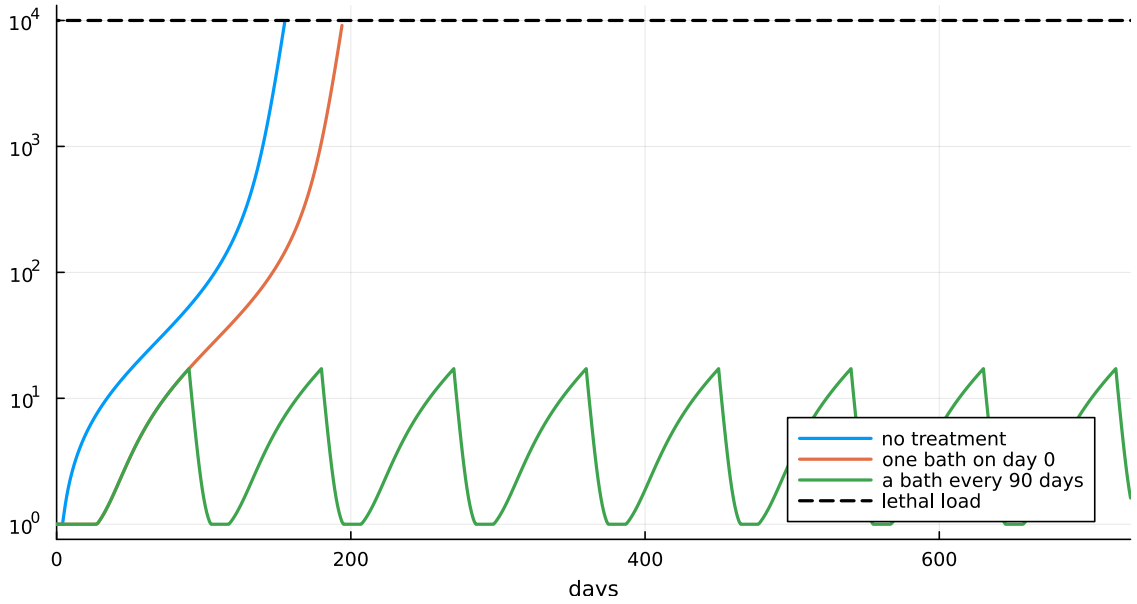
Figure 8: Bacteria on frog i: set to 1 by a bath, lost at rate δ, exchanged with other frogs, and raising the clearance of sporangia (dashed).

```
one = baths()
step!("Bd frog", "treatment", "+ bacteria on each frog, fitted decay; one
 ↪  bath", alive(r1),
    one.alive[end] / N, "die-off delayed by about six weeks")
```

```
(step = 5, untreated = 0.0, treated = 0.0)
```

Step 6. Repeated baths

```
untreated, every90 = baths(coverage = 0.0), baths(dose_days = 0:90:729)
plot(untreated.t, max.(untreated.mean_load, 1.0); yscale = :log10, label =
 ↪  "no treatment",
    xlabel = "days", ylabel = "mean Bd load per live frog", legend =
 ↪  :bottomright, xlims = (0, 730))
plot!(one.t, max.(one.mean_load, 1.0); label = "one bath on day 0")
plot!(every90.t, max.(every90.mean_load, 1.0); label = "a bath every 90
 ↪  days")
hline!([pb.Smax]; color = :black, linestyle = :dash, label = "lethal load")
```




```
first_death(r) = any(>(0), r.death_day) ? minimum(filter(>(0), r.death_day))
 ↪  : 0
println("| Baths | Alive after 2 years (of 80) | First death (day) |")
println("|---|---|---|")
for (label, r) in (("none", untreated), ("one, day 0", one), ("every 240
 ↪  days", baths(dose_days = 0:240:729)),
        ("every 150 days", baths(dose_days = 0:150:729)), ("every 90 days",
 ↪  every90),
        ("every 90 days, 10% of frogs", baths(coverage = 0.1, dose_days =
 ↪  0:90:729)),
        ("every 90 days, 10% of frogs, no exchange", baths(coverage = 0.1, x
 ↪  = 0.0, dose_days = 0:90:729)),
        ("one, day 150 (loads already high)", baths(dose_days = [150])))
    @printf("| %s | %d | %s |\n", label, r.alive[end], first_death(r) == 0 ?
     ↪  "none" : string(first_death(r)))
end
```

Baths	Alive after 2 years (of 80)	First death (day)
none	0	156
one, day 0	0	195
every 240 days	0	195
every 150 days	80	none
every 90 days	80	none
every 90 days, 10% of frogs	80	none
every 90 days, 10% of frogs, no exchange	8	184
one, day 150 (loads already high)	0	157

```
step!("Bd frog", "treatment", "+ a bath every 90 days", alive(r1),
 ↪  every90.alive[end] / N,
    "all survive: each bath resets the slow build-up of load")
```

```
(step = 6, untreated = 0.0, treated = 1.0)
```

This is the one system where bacteria that last only weeks could still work. Load takes months to build; a short burst of extra clearance knocks it back, and the build-up starts again. Because frogs exchange bacteria, bathing a small part of the population protects the rest.

Step 7. Chance

The same model with every sporangium on every frog, and every zoospore in the pool, a whole number, and every event random (`StochasticFrogProblem`); 20 runs of each case.

Time moves in steps of $dt = 0.05$ days. In each step the random events have the rates of the equations, so on average they follow them:

Event in a step	Number drawn	Term in the equations
new sporangia on frog i	Poisson with mean $\left(\frac{\nu\gamma}{V}Z + \eta\nu f S_i\right) dt$	first two terms of dS_i/dt
sporangia lost from frog i	Binomial(S_i, $1 - e^{-(\sigma + c_i/(h+S_i))\,dt}$)	$\sigma S_i + c_i S_i/(h + S_i)$
zoospores released by frog i	Poisson with mean $\eta(1-f)S_i\,dt$	$\eta(1-f)\sum_j S_j$
zoospores lost from the pool	Binomial(Z, $1 - e^{-(\mu + \gamma N_{\text{alive}}/V)\,dt}$)	$\mu Z + (\gamma/V)N_{\text{alive}}Z$
zoospores from outside	Poisson with mean $\varepsilon\,dt$	ε

A frog dies as soon as S_i reaches $S_{\max}$. The bacteria B_i stay continuous and follow the step-5 equation, solved exactly over each step.

```
bath_tr = T(abundance_half_life = 5.0, protective_fold = 100.0, direct =
 ↪  0.1)
chanceC(; tr = bath_tr, kw...) = [solve(StochasticFrogProblem(pb, tr; c,
 ↪  c_treated = 3c, h = hsat, N, ε,
    days = 730, coverage = 0.75, kw...); rng = Xoshiro(s)) for s in 1:20]
aliveC(rs) = sum(r.alive[end] for r in rs) / (N * length(rs))
firstC(rs) = (v = sort([first_death(r) for r in rs if first_death(r) > 0]);
 ↪  isempty(v) ? "none" : string(v[cld(length(v), 2)]))
cases = [("no treatment", chanceC(coverage = 0.0), untreated),
    ("bath every 90 days", chanceC(dose_days = 0:90:729), every90),
    ("bath every 150 days", chanceC(dose_days = 0:150:729), baths(dose_days
 ↪  = 0:150:729)),
    ("bath every 90 days, 10% of frogs", chanceC(coverage = 0.1, dose_days =
 ↪  0:90:729),
        baths(coverage = 0.1, dose_days = 0:90:729))]
nothing
```

```
println("| Baths | Equations: alive after 2 years | Chance: alive (mean of
 ↪  20 runs) | Chance: runs with no deaths | Chance: median first death |")
println("|---|---|---|---|---|")
for (label, rs, det) in cases
    @printf("| %s | %d | %.0f | %d of 20 | %s |\n", label, det.alive[end],
     ↪  aliveC(rs) * N,
        count(r -> r.alive[end] == N, rs), firstC(rs))
end
```

Baths	Equations: alive after 2 years	Chance: alive (mean of 20 runs)	Chance: runs with no deaths	Chance: median first death
no treatment	0	0	0 of 20	114
bath every 90 days	80	80	20 of 20	none
bath every 150 days	80	0	0 of 20	154
bath every 90 days, 10% of frogs	80	24	6 of 20	482

```
step!("Bd frog", "chance", "+ chance: whole sporangia and zoospores",
 ↪  aliveC(cases[1][2]), aliveC(cases[2][2]),
    "baths every 90 days still protect; less frequent or narrower bathing
     ↪  now fails")
```

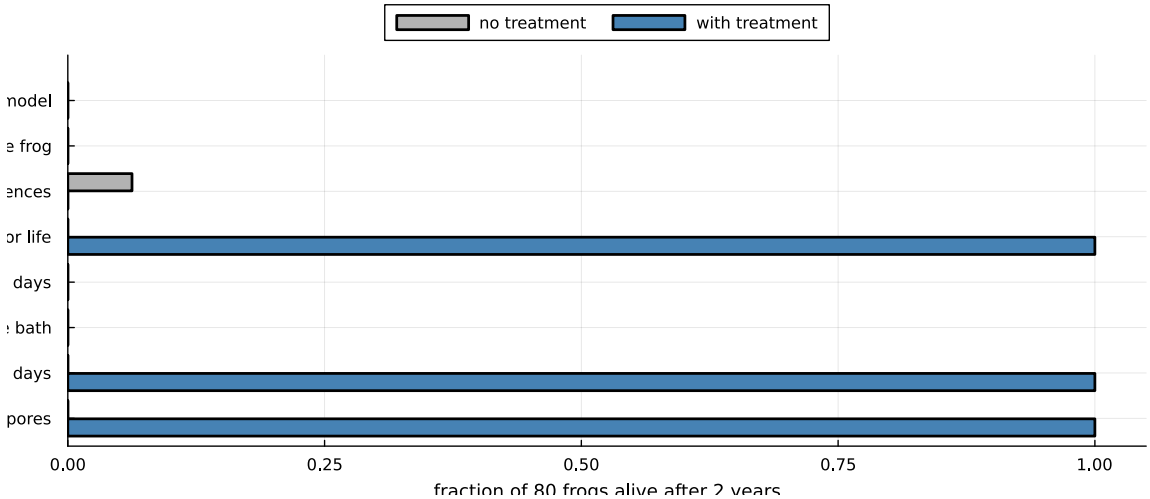
```
(step = 7, untreated = 0.0, treated = 1.0)
```

Chance makes the frog system harder to protect. Untreated, frogs die about six weeks earlier than in the equations: a few frogs whose loads happen to run high feed the shared pool and tip the rest. Baths every 90 days still protect every frog, but baths every 150 days, which were enough in the equations, now fail, and bathing 10% of frogs works in only some runs.

Two cautions. The slow build-up is for a population that starts clean. In the field trial the pond was in the middle of an outbreak, treated frogs were surrounded by heavily infected ones, and their loads were back within two months; the last row of the table is the model's version of treating too late. And the size of the clearance effect of the bacteria is assumed, not measured.

Part C summary

```
ladder_plot("Bd frog", "fraction of 80 frogs alive after 2 years")
```

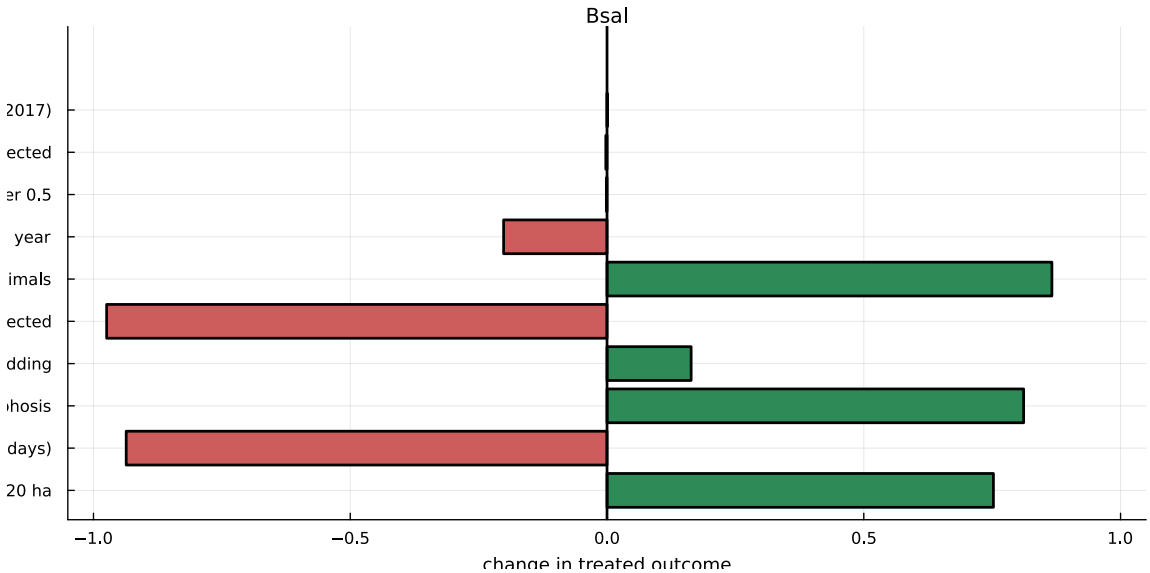


How much did each step move the answer?

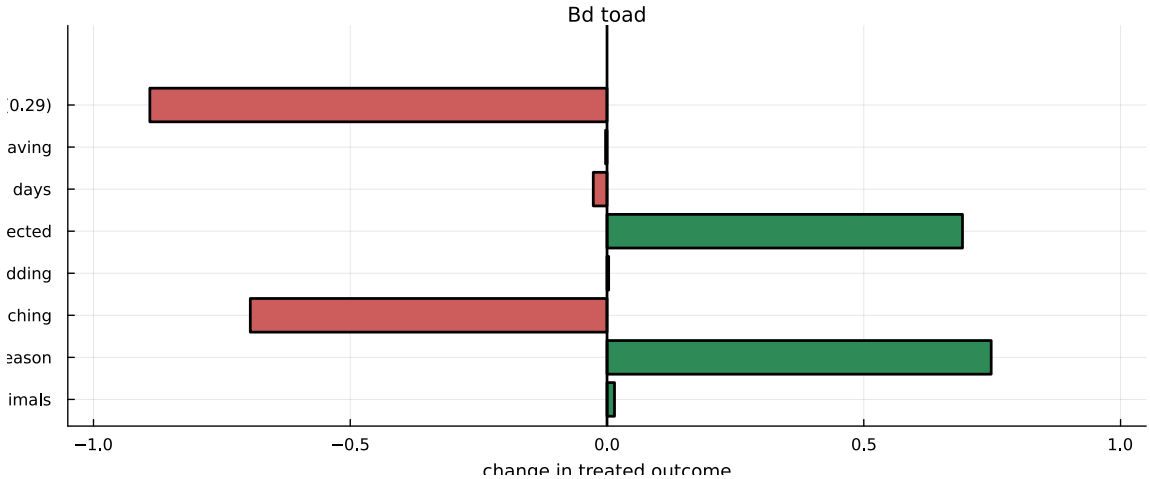
The change in the treated outcome at each step, for each system. Bars to the right are steps that made treatment look better; bars to the left, worse.

```
function change_plot(system)
    rows = filter(r -> r.system == system && !isnan(r.treated), ladder)
    d = [rows[i].treated - rows[i - 1].treated for i in 2:length(rows)]
    labels = ["$(rows[i].n). $(rows[i].added)" for i in 2:length(rows)]
    bar(d; orientation = :h, yticks = (1:length(d), labels), yflip = true,
     ↪  legend = false,
```

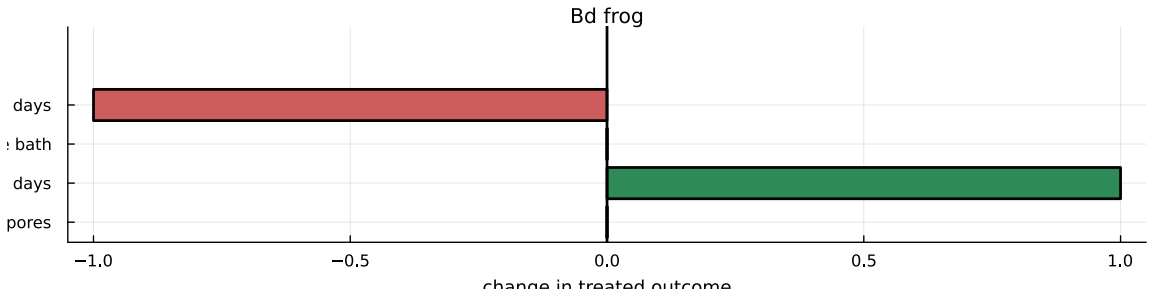
```
        xlims = (-1.05, 1.05), xlabel = "change in treated outcome", title
↪   = system,
        color = [x >= 0 ? :seagreen : :indianred for x in d], left_margin =
↪   6Plots.mm,
        size = (840, 70 + 34 * length(d)))
    vline!([0.0]; color = :black)
end
change_plot("Bsal")
```




```
change_plot("Bd toad")
```




```
change_plot("Bd frog")
```



Large bars point both ways in every system. No single addition dominates, and the sign of a step depends on the system: the observed mechanism (survive, still infected) is the largest negative step for Bsal and a large positive step for toads.

Does the effect of one piece of realism depend on the others?

A ladder adds things in one order. To check that the order is not driving the conclusions, four pieces of treatment realism are switched on and off in every combination for Bsal, on the full host model of step 3:

- mechanism: blocks infection (0.99), or survive but stay infected (0.99);
- duration: lasts for life, or fitted decay (half-life 5 days);
- spread: none, or between animals at 50 per year;
- stages: adults once, or both stages every quarter with the bacteria kept through metamorphosis.

```
levels = (
    mechanism = ["blocks", "survive"],
    duration = ["lasting", "decays"],
    spread = ["none", "spreads"],
    stages = ["adults", "both"],
)
function factorial_run(mech, dur, spr, stg)
    tr = T(; (mech == "blocks" ? (efficacy = 0.99,) : (tolerance =
 ↪  0.99,))...,
        (dur == "decays" ? (abundance_half_life = 5 / 365, protective_fold =
 ↪  100.0) : (;))...,
        direct = spr == "spreads" ? 50.0 : 0.0)
    kw = stg == "both" ? (treat = :both, treat_times = quarterly, retention
 ↪  = 1.0) : (;)
    return lar(stage, tr; coverage = 1.0, kw...).mean_adults
end
design = [(; mech, dur, spr, stg, outcome = factorial_run(mech, dur, spr,
 ↪  stg))
          for mech in levels.mechanism, dur in levels.duration, spr in
 ↪  levels.spread, stg in levels.stages]
```

```
design = sort(vec(design); by = r -> -r.outcome)
nothing
```

```
println("| Mechanism | Duration | Spread | Stages treated | Mean adults |")
println("|---|---|---|---|---|")
for r in design
    @printf("| %s | %s | %s | %s | %.2f |\n", r.mech, r.dur, r.spr, r.stg,
     ↪  r.outcome)
end
```

Mechanism	Duration	Spread	Stages treated	Mean adults
blocks	lasting	spreads	both	1.00
blocks	lasting	none	both	1.00
blocks	lasting	spreads	adults	1.00
blocks	lasting	none	adults	0.33
survive	lasting	spreads	both	0.27
survive	lasting	none	both	0.22
survive	lasting	spreads	adults	0.20
survive	lasting	none	adults	0.13
blocks	decays	spreads	both	0.07
blocks	decays	none	both	0.06
blocks	decays	spreads	adults	0.05
blocks	decays	none	adults	0.05
survive	decays	spreads	adults	0.04
survive	decays	none	adults	0.04
survive	decays	none	both	0.03
survive	decays	spreads	both	0.03

The average effect of switching each piece on, and how much that effect varies with the other three:

```
function effect(field, on)
    diffs = Float64[]
    for r in design
        getfield(r, field) == on || continue
        partner = only(filter(q -> all(getfield(q, f) == getfield(r, f) for
 ↪  f in (:mech, :dur, :spr, :stg) if f != field) &&
            getfield(q, field) != on, design))
        push!(diffs, r.outcome - partner.outcome)
    end
    return diffs
end
```

```
println("| Switched on | Mean change | Smallest | Largest |")
println("|---|---|---|---|")
for (field, on, label) in ((:mech, "survive", "survive instead of block"),
 ↪  (:dur, "decays", "fitted decay"),
        (:spr, "spreads", "spread between animals"), (:stg, "both", "treat
 ↪  both stages"))
    d = effect(field, on)
    @printf("| %s | %+.2f | %+.2f | %+.2f |\n", label, sum(d) / length(d),
     ↪  minimum(d), maximum(d))
end
```

Switched on	Mean change	Smallest	Largest
survive instead of block	-0.32	-0.80	-0.00
fitted decay	-0.47	-0.95	-0.09
spread between animals	+0.10	-0.00	+0.67
treat both stages	+0.10	-0.01	+0.67

The ranges are wide: the effect of each piece depends strongly on the others. Spread, for example, matters a great deal for a lasting, infection-blocking treatment given to adults, and hardly at all once the bacteria decay in days. (In the model, animals that acquire bacteria from others share the dosing clock of the treated ones, so spread cannot outlast the decay.) A piece of realism therefore cannot be judged by adding it to a simple model alone.

All steps

```
fmt(x) = isnan(x) ? "" : @sprintf("%.2f", x)
println("| System | Step | Phase | Added | No treatment | With treatment |
 ↪  What changed |")
println("|---|---|---|---|---|---|---|")
for r in ladder
    println("| ", join((r.system, r.n, r.phase, r.added, fmt(r.untreated),
     ↪  fmt(r.treated), r.note), " | "), " |")
end
```

System	Step	Phase	Added	No treatment	With treatment	What changed
Bsal	0	host	susceptible and infectious adults	0.00	0.34	collapse; treated adults live but are not replaced
Bsal	1	host	+ latent period (Schmidt 2017)	0.00	0.34	almost no change: a one-day delay against a one-week illness
Bsal	2	host	+ larval stage, not infected	0.04	0.34	untreated population no longer vanishes: larvae are a refuge
Bsal	3	host	+ larvae infected, carry-over 0.5	0.04	0.33	small change at this carry-over
Bsal	4	treatment	+ protection lost, half-life 1 year	0.04	0.13	collapse only delayed
Bsal	5	treatment	+ bacteria spread between animals	0.04	1.00	full protection: the microbiome maintains itself
Bsal	6	treatment	mechanism as in trials: survive, still infected	0.04	0.02	fails; survivors shed for longer, so transmission rises
Bsal	7	treatment	+ lower load (99%): also less shedding	0.04	0.19	partial; survivors remain infected

System	Step	Phase	Added	No treatment	With treatment	What changed
Bsal	8	treatment	treat both stages, kept through metamor-phosis	0.04	1.00	full protection with no spread at all
Bsal	9	treatment	+ fitted decay (half-life 5 days)	0.04	0.06	fails: protection lasts about a month
Bsal	10	chance	+ chance: whole animals, 20 ha	0.08	0.82	the failed treatment of step 9 now keeps most of the population
Bd toad	0	host	Mitchell 2008 pond model	0.00	0.98	untreated: toads and Bd both die out; treated: toads saved, Bd gone
Bd toad	1	host	+ infected meta-morphs can survive (0.29)	0.05	0.09	toads no longer die out, so nor does Bd; treatment now protects poorly
Bd toad	2	host	+ meta-morphs exposed for a week on leaving	0.05	0.08	little further change
Bd toad	3	host	+ adults carry Bd between years	0.04		untreated only; little change in this parameter set

System	Step	Phase	Added	No treatment	With treatment	What changed
Bd toad	4	treatment	+ protection lost, half-life 18 days	0.05	0.06	slightly worse
Bd toad	5	treatment	mecha- nism as in trials: survive, still infected	0.05	0.75	much better than blocking here; Bd persists
Bd toad	6	treatment	+ lower load (70%): also less shedding	0.05	0.75	little changed: untreated tadpoles and the pond keep Bd going
Bd toad	7	treatment	+ fitted decay, one dose at hatching	0.05	0.06	protection gone before the season ends
Bd toad	8	treatment	+ five doses per season	0.05	0.80	most of the benefit restored
Bd toad	9	chance	+ chance: whole animals	0.07	0.82	little change for this model; large changes elsewhere (see text)
Bd frog	0	host	Briggs 2010 load model	0.00		all die: 80 frogs is above the threshold of 23
Bd frog	1	host	+ saturating clearance by the frog	0.00		threshold rises from 23 to 68 frogs; 80 still all die

System	Step	Phase	Added	No treatment	With treatment	What changed
Bd frog	2	host	+ frogs differ in their defences	0.06		a few of the best-defended survive
Bd frog	3	treatment	treated class: 40% of frogs, lasts for life	0.00	1.00	all survive, including the untreated: protection is shared
Bd frog	4	treatment	+ protection lost, half-life 30 days	0.00	0.00	die-off delayed by about three months, not prevented
Bd frog	5	treatment	+ bacteria on each frog, fitted decay; one bath	0.00	0.00	die-off delayed by about six weeks
Bd frog	6	treatment	+ a bath every 90 days	0.00	1.00	all survive: each bath resets the slow build-up of load
Bd frog	7	chance	+ chance: whole sporangia and zoospores	0.00	1.00	baths every 90 days still protect; less frequent or narrower bathing now fails

What the exercise shows

1. The answer changed at almost every step, and reversed more than once. For Bsal, a model stopped at step 5 says a self-spreading microbiome is essential; at step 8, that spread is unnecessary; at step 9, that neither is achievable with bacteria that last a few weeks.

2. Host biology can move the answer as much as treatment biology, but not everywhere. A larval stage changed the untreated fate of the salamanders. One survival probability for infected toad metamorphs reversed both the untreated outcome and the value of the reference treatment. The second vignette tests this like for like in a single model and finds that it depends on the regime: for Bsal at the published host density, six pieces of host biology barely change the outcome, and near the invasion threshold they change it a great deal. Its stochastic version also shows that chance can matter more than any of them: with whole animals, blocking infection in every adult on arrival nearly always eliminates Bsal, where the equations of step 1 here leave about a third of the population.
3. How protection works matters more than how strong it is, and its effect has no fixed sign. Letting animals survive infection is the worst change for Bsal and one of the best for toads.
4. Effects interact. The factorial shows that the value of spread, of treating larvae or of a lasting treatment each depends on the other pieces.
5. The systems differ for structural reasons. A toad cohort passes through a short infected stage once a year and can be re-treated. A salamander must be protected for life against a pathogen that kills in a week. In frogs, where disease follows a load that builds slowly, short-lived bacteria can still work if they are re-applied.
6. Chance is a piece of realism, and it changed the answer in every system, in different directions. For Bsal it turned a failed treatment into a successful one, because Bsal arriving among treated animals is lost before it can establish. For toads it overturned the published extinction of both host and pathogen. For frogs it made treatment harder, because random high loads tip the population sooner.

What is not in the models

- In Parts A and B bacterial abundance is a clock since the last dose, not a state of each animal. Only the frog model of Part C tracks it per animal.
- The fold-drop in abundance at which protection is lost has not been measured, nor has the effect of the bacteria on clearance, nor whether applied bacteria survive metamorphosis.
- The Bsal model has no recovery from infection and no years between metamorphosis and maturity. The toad model has thresholds standing in for extinction, and infected adult toads are not yet combined with treatment. The frog treatment steps use identical frogs.
- The three systems do not share one model structure, so Part B starts from a more complex model than Part A. The second vignette uses one nested model, for the salamander system, in which every piece of biology is a switch.
- No fade-out rule is applied in Part A, so densities can fall to meaningless values and recover. The second vignette shows what such a rule changes.
- The stochastic steps use one introduced infected animal (Part A) or the published introduction (Parts B and C), one population size, and time steps of hours to a day. In Part A the area is 20 hectares; whether the host is lost depends on it (see the second vignette).

References

- Briggs CJ, Knapp RA, Vredenburg VT (2010) Enzootic and epizootic dynamics of the chytrid fungal pathogen of amphibians. *Proceedings of the National Academy of Sciences* 107:9695–9700. doi:10.1073/pnas.0912886107
- Harris RN, Brucker RM, Walke JB, Becker MH, et al. (2009) Skin microbes on frogs prevent morbidity and mortality caused by a lethal skin fungus. *The ISME Journal* 3:818–824. doi:10.1038/ismej.2009.27
- Knapp RA, Joseph MB, Smith TC, Hegeman EE, et al. (2022) Effectiveness of antifungal treatments during chytridiomycosis epizootics in populations of an endangered frog. *PeerJ* 10:e12712. doi:10.7717/peerj.12712
- Madison JD, Osborne OG, Ellison A, Garvey Griffith CN, et al. (2025) Probiotic colonization of *Xenopus laevis* skin causes short-term changes in skin microbiomes and gene expression. *Infection and Immunity* 93:e00569-24. doi:10.1128/iai.00569-24
- Mitchell KM, Churcher TS, Garner TWJ, Fisher MC (2008) Persistence of the emerging pathogen *Batrachochytrium dendrobatidis* outside the amphibian host greatly increases the probability of host extinction. *Proceedings of the Royal Society B* 275:329–334. doi:10.1098/rspb.2007.1356
- Schmidt BR, Bozzuto C, Lötters S, Steinfartz S (2017) Dynamics of host populations affected by the emerging fungal pathogen *Batrachochytrium salamandrivorans*. *Royal Society Open Science* 4:160801. doi:10.1098/rsos.160801
- Smith D, O'Brien D, Hall J, Sergeant C, et al. (2022) Challenging a host–pathogen paradigm: susceptibility to chytridiomycosis is decoupled from genetic erosion. *Journal of Evolutionary Biology* 35:589–598. doi:10.1111/jeb.13987
- Wang Y, Verbrugghe E, Meuris L, Chiers K, et al. (2021) Epidermal galactose spurs chytrid virulence and predicts amphibian colonization. *Nature Communications* 12:5788. doi:10.1038/s41467-021-26127-9